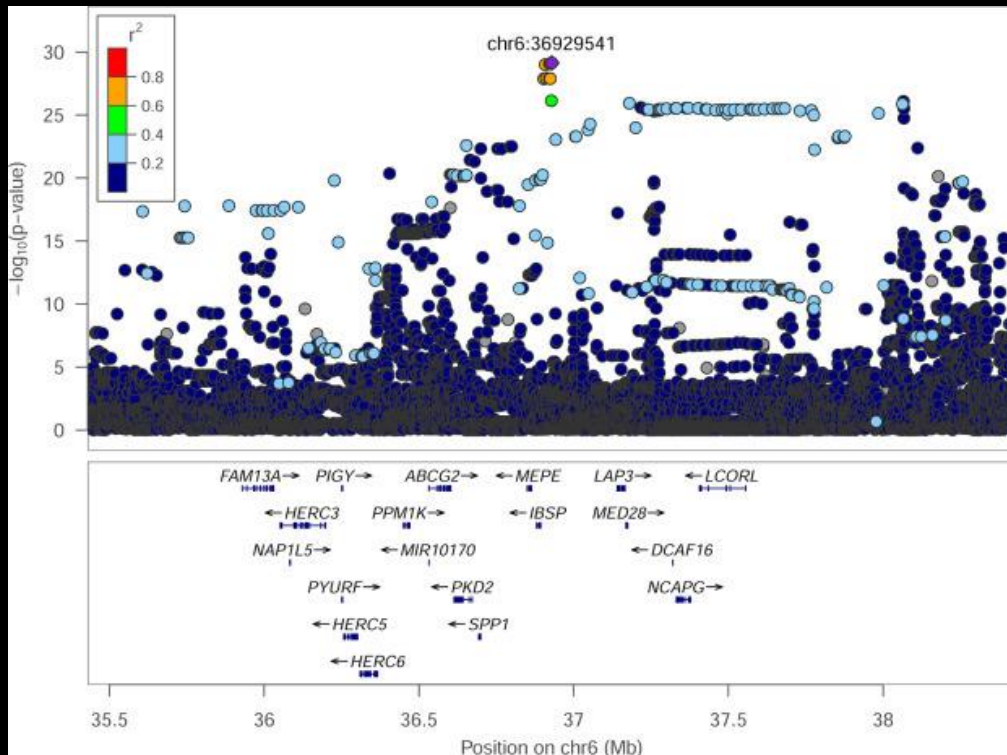


Fine-mapping and examples of data integration

HUGEN 2073

Jon Chernus

Why we need fine-mapping

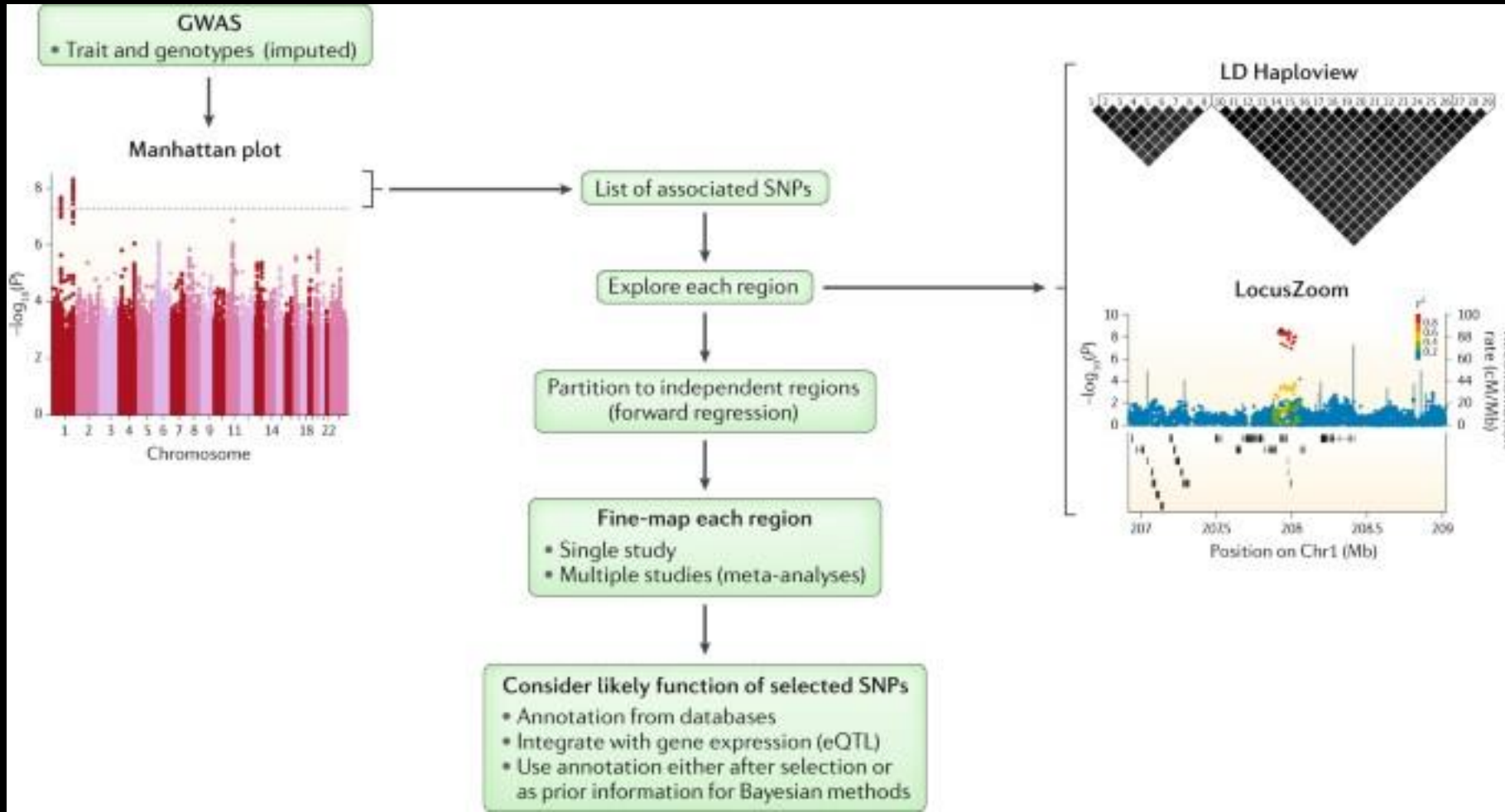


GWAS

- Identifies a set of correlated variants
- One (or more) of these SNPs is causal – but which?
- Peaks are broad - dozens of significant SNPs
- The whole LD block is a candidate

Fine-mapping = shrinking the set of candidates

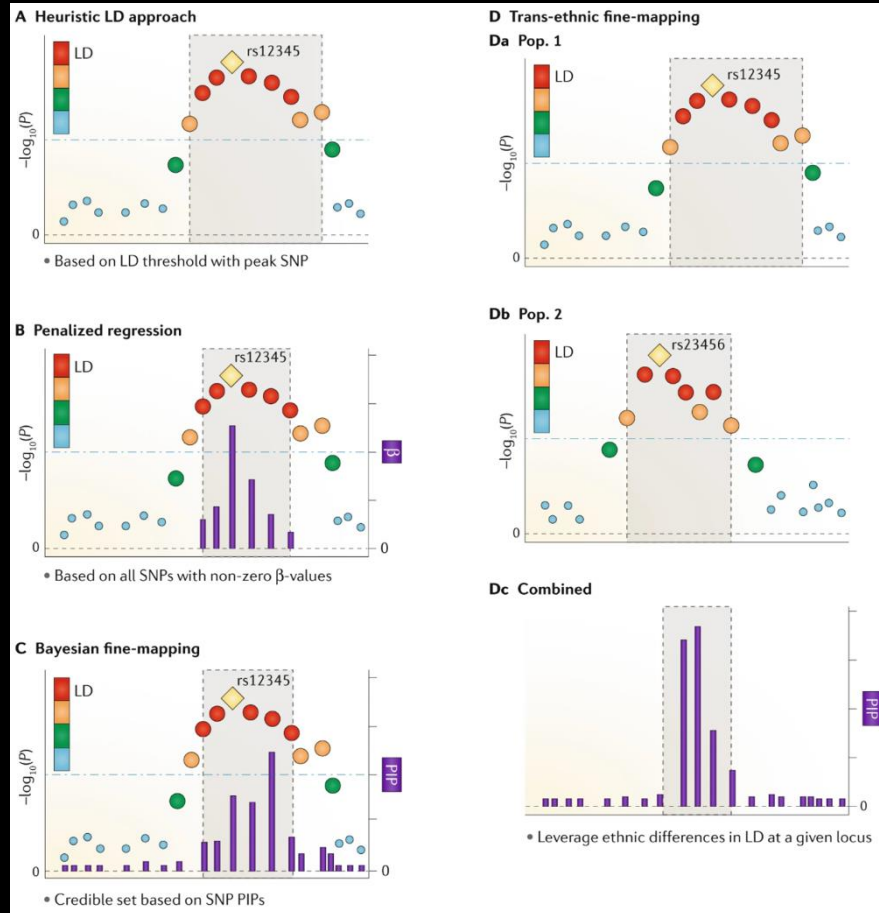
Fine-mapping



GWAS: What (set of) variants are associated with my trait?

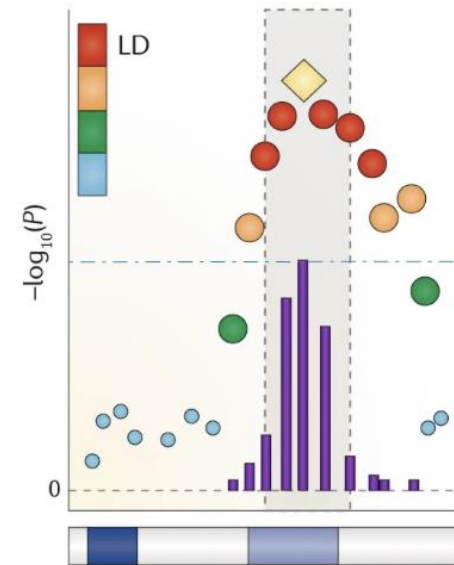
Fine-mapping: “seeks to determine the genetic variant (or variants) responsible for complex traits, given evidence of an association of a genomic region with a trait and assuming at least one causal variant exists.”

Some fine-mapping strategies



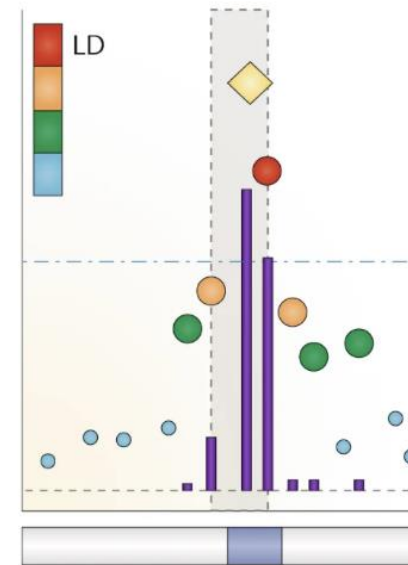
E Multi-region fine-mapping with annotation

Ea Locus 1

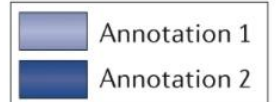
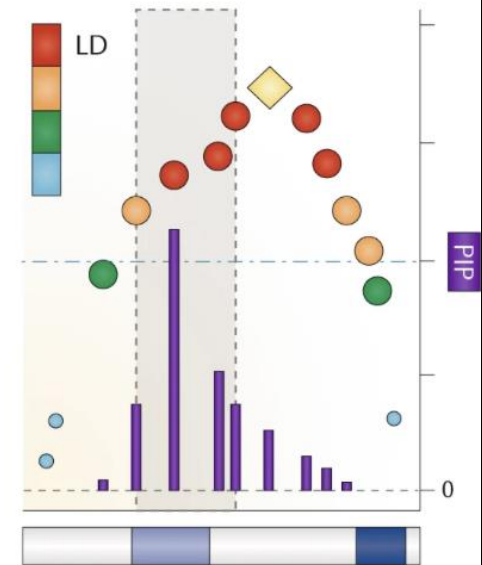


- Leverage annotation enrichment
- Borrow information across loci

Eb Locus 2



Ec Locus 3



Bayesian fine-mapping under the hood

- Input: summary stats for a locus + LD
- Consider many possible causal configurations of SNPs
- Assign each a small prior probability of being true
- For each configuration, How well does it explain the observed association patterns, given LD?
- Use Bayes' rule to compute posterior probability of each configuration
- Sum over configurations containing SNP $j \rightarrow$ PIP for that SNP

Some fine-mapping tools

“**Posterior inclusion probability(PIP)** = The marginal probability that a single-nucleotide polymorphism (SNP) is included in any causal model, conditional on the observed data, thereby providing weight of evidence that a SNP should be included as potentially causative.

Bayesian methods can be used to determine the **credible set**, the minimum set of SNPs that contains all causal SNPs with probability α . When assuming only one causal SNP, α is the sum of PIPs for SNPs in a set. This means that an α credible set is equivalent to ranking SNPs from largest to smallest PIPs⁵⁰ and taking the cumulative sum of PIPs until it is at least α .”

Another, newer popular tool:

- SusieR (R package)

Table 1 Commonly used Bayesian fine-mapping software

From: [From genome-wide associations to candidate causal variants by statistical fine-mapping](#)

Software	Trait type ^a	Input covariates ^b	Uses summary statistics?	Maximum number of causal variants ^c	Input annotation?	Causal search	Main output	Refs
BIMBAM v1.0	qt and binary	No	No	Fixed	No	Exhaustive	Bayes factor	113 , 114
mvBIMBAM v1.0.0	mqt	No	Yes	1	No	Exhaustive	Bayes factor	115 , 116
SNPTEST v2.5.4-beta3	qt, binary, mqt and multinomial	No	No	1	No	Exhaustive	Bayes factor	117
piMASS v0.9	qt and binary	No	No	Computed	No	MCMC	Bayes factor and PIP	45
BVS v4.12.1	Binary	Yes	No	Computed	Yes	MCMC	Bayes factor and PIP	97 , 118 , 119
FM-QTL	qt	No	No	Computed	Yes	MCMC	Bayes factor and PIP	96
DAP v1.0.0	qt	Yes	Yes	1, fixed and computed	Yes	Exhaustive	Bayes factor and PIP	52
Fine-mapping	Multinomial	Yes	No	Computed	No	Greedy	PIP	30
Trinculo	Multinomial	Yes	No	Computed	No	Greedy	Bayes factor and PIP	30 , 120
BayesFM	Binary	Yes	No	20	No	MCMC	PIP	30
ABF	qt and binary ^d	Yes	Yes	1	No	Exhaustive	Bayes factor	121
fgwas v0.3.6	qt and binary ^d	No	Yes	1	Yes	Exhaustive	Bayes factor and PIP	95
CAVIAR/eCAVIAR	qt and binary ^d	No	Yes	Fixed	No	Exhaustive	ρ probability confidence set and PIP	46 , 101
PAINTOR v3.0	qt, binary ^d and mqt	No	Yes	Fixed and computed	Yes	Exhaustive and MCMC	Bayes factor and PIP	54 , 57 , 71
CAVIARBF v0.2.1	qt and binary ^d	No	Yes	Fixed	Yes	Exhaustive	Bayes factor and PIP	47 , 94
FINEMAP v1.1	qt and binary ^d	No	Yes	Fixed	No	Shotgun stochastic search	Bayes factor and PIP	53
JAM in R2BGLIMS v0.1	qt and binary ^d	No	Yes	Fixed and computed	No	Exhaustive and MCMC	Bayes factor and PIP	55

Colocalization

Often we have two signals at the same locus, e.g.:

- GWAS signal for a disease
- eQTL signal for a gene

That could mean

- the same SNP causes both (shared biology)
- different SNPs in LD (potentially misleading)

Colocalization = distinguishing these

Colocalization

“One approach to understanding the biological basis of these associations has been to explore whether GWAS signals from intermediate cellular phenotypes, in particular gene expression, are located in the same loci (“colocalise”) and are potentially mediating the disease signals. However, it is not clear how to assess whether the same variants are responsible for the two GWAS signals or whether it is distinct causal variants close to each other. In this paper, we describe a statistical method that can use simply single variant summary statistics to test for colocalisation of GWAS signals.” - COLOC

Coloc is a Bayesian method which, for two traits, integrates evidence over all variants at locus to evaluate the following hypotheses:

H0: No association with either trait

H1: Association with trait 1, not with trait 2

H2: Association with trait 2, not with trait 1

H3: Association with trait 1 and trait 2, two independent SNPs

H4: Association with trait 1 and trait 2, one shared SNP

Some other tools:

- HyPrColoc

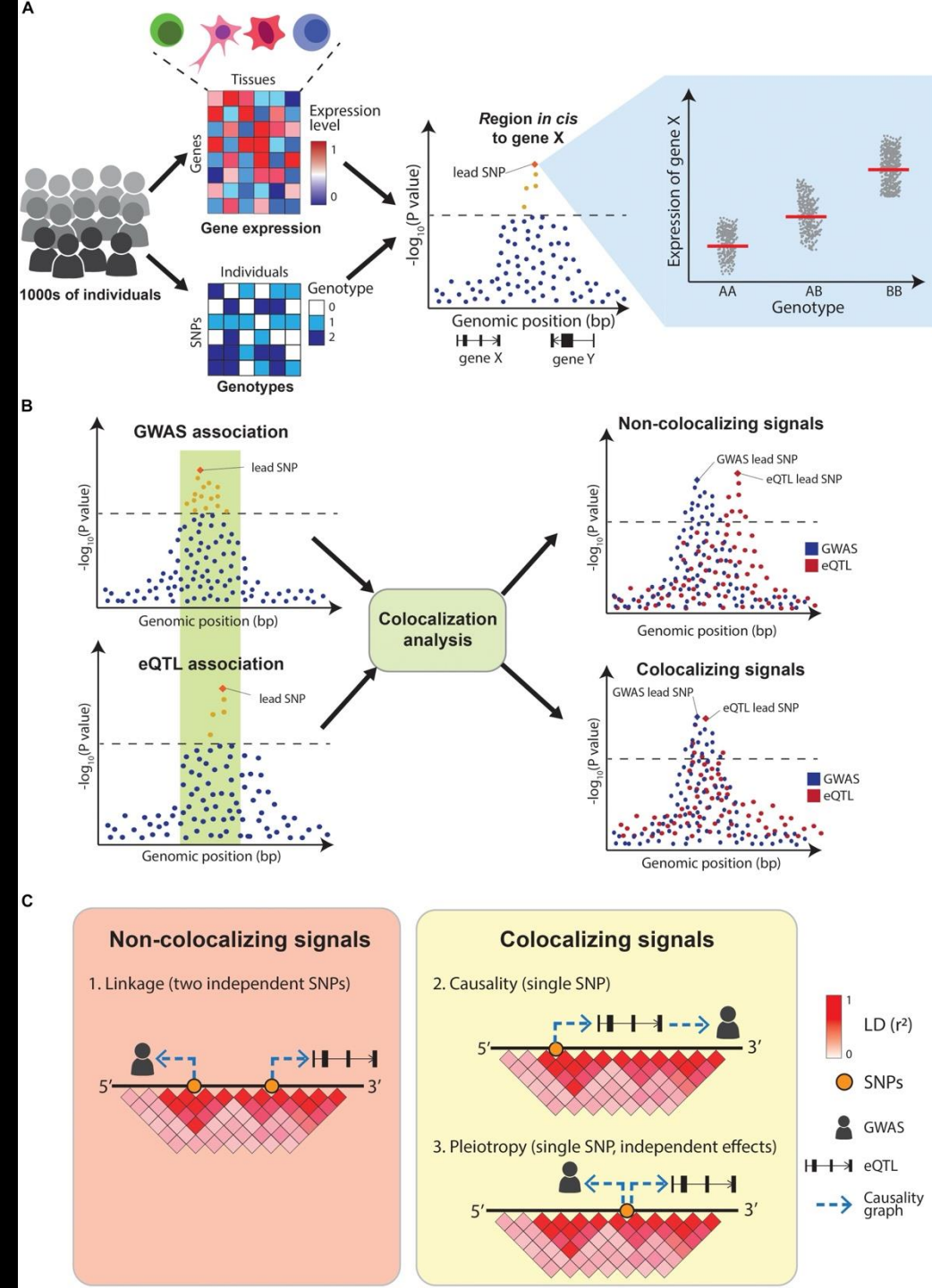
Colocalization

“Overlapping eQTL and GWAS signals can be explained by three possible scenarios:

- (1) two independent causal SNPs in LD with each other (linkage)
- (2) a single-causal SNP which affects the trait by modulating the expression of a gene (causality), or
- (3) a single-causal SNP with independent effects on trait and gene expression (pleiotropy).

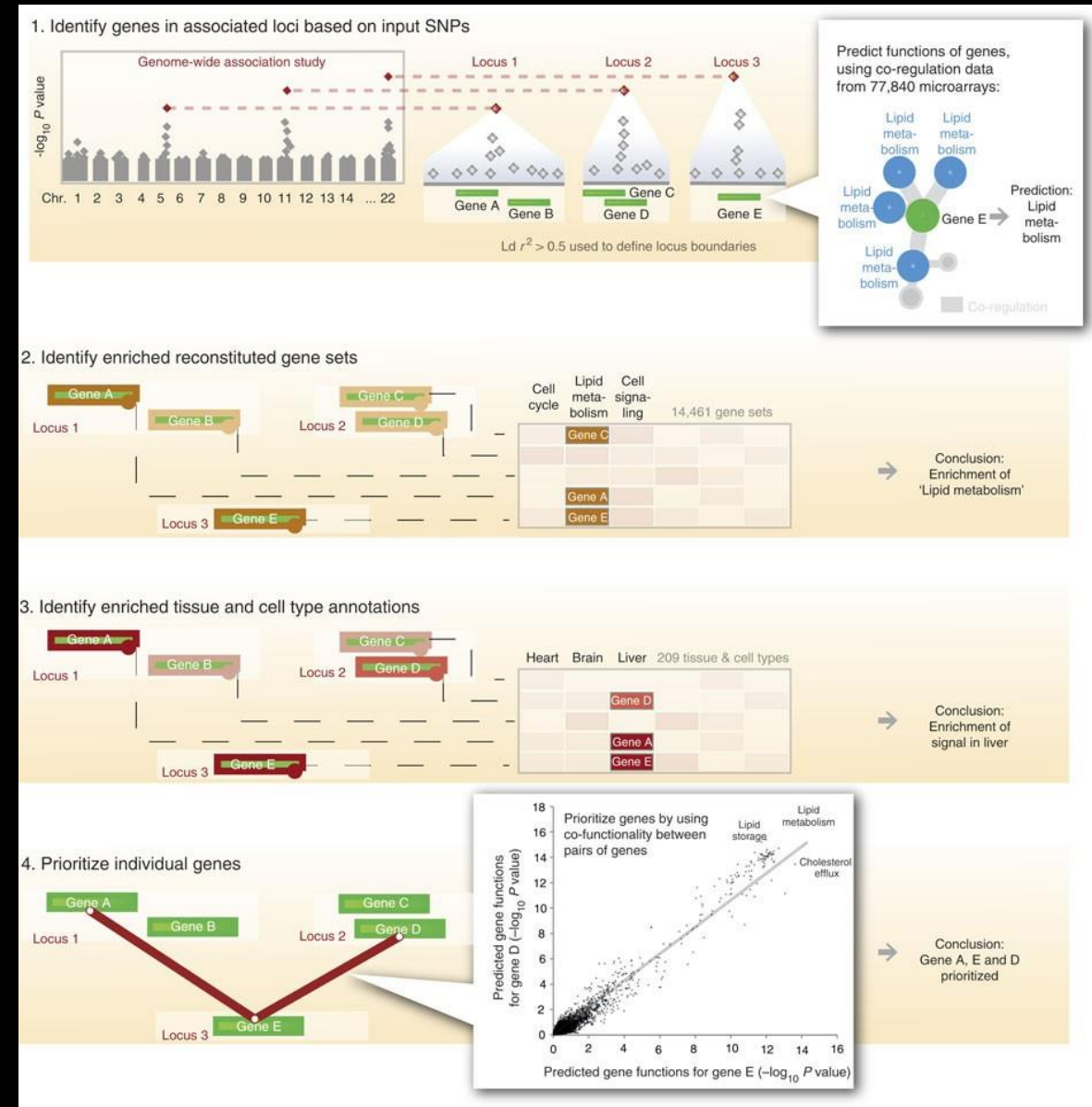
Distinguishing between these scenarios is crucial to appropriately interpret GWAS results”

Colocalization doesn't distinguish (2) and (3)!



Another tool: DEPICT

DEPICT is designed to identify likely causal genes, functional or phenotypic gene sets that are enriched in genes within associated loci, and tissues or cell types that are implicated by the associated loci. DEPICT takes as input a set of trait-associated SNPs and uses them to identify independently associated loci that may comprise up to several genes. DEPICT uses co-regulation data from 77,840 microarrays to predict genes' biological functions across 14,461 gene sets representing a wide spectrum of biological annotations and to construct 14,461 'reconstituted' gene sets. DEPICT then uses this information to identify reconstituted gene sets that enrich for genes in the associated loci, and to prioritize genes at associated loci, by identifying genes in different loci that have similar predicted functions. Finally, DEPICT relies on 37,427 human gene expression microarrays to assess whether genes in associated loci are highly expressed in any of 209 tissue/cell type annotations.



GWAS examples

How are the data integration resources and tools we've discussed used in real publications?

We'll look at examples and try to characterize a few GWA studies based on their "downstream" analyses.

GWAS #1

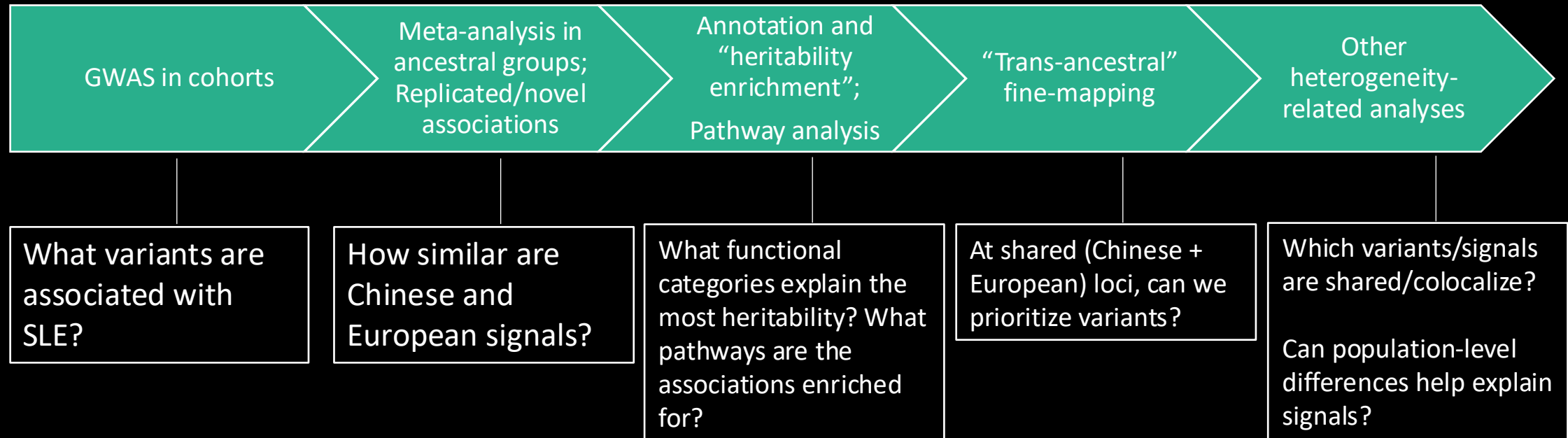
- Phenotype: systemic lupus erythematosus (SLE)
- Predominantly studied so far in Europeans
- This study performs GWAS in Chinese individuals (largely Han) and re-analyzes existing GWAS cohorts
- Known population-level differences: non-Europeans have
 - earlier age-at-onset
 - higher prevalence
 - higher risk of renal disease

Wang, Y. F., Zhang, Y., Lin, Z., Zhang, H., Wang, T. Y., Cao, Y., Morris, D. L., Sheng, Y., Yin, X., Zhong, S. L., Gu, X., Lei, Y., He, J., Wu, Q., Shen, J. J., Yang, J., Lam, T. H., Lin, J. H., Mai, Z. M., Guo, M., ... Yang, W. (2021). Identification of 38 novel loci for systemic lupus erythematosus and genetic heterogeneity between ancestral groups. *Nature communications*, 12(1), 772. <https://doi.org/10.1038/s41467-021-21049-y>

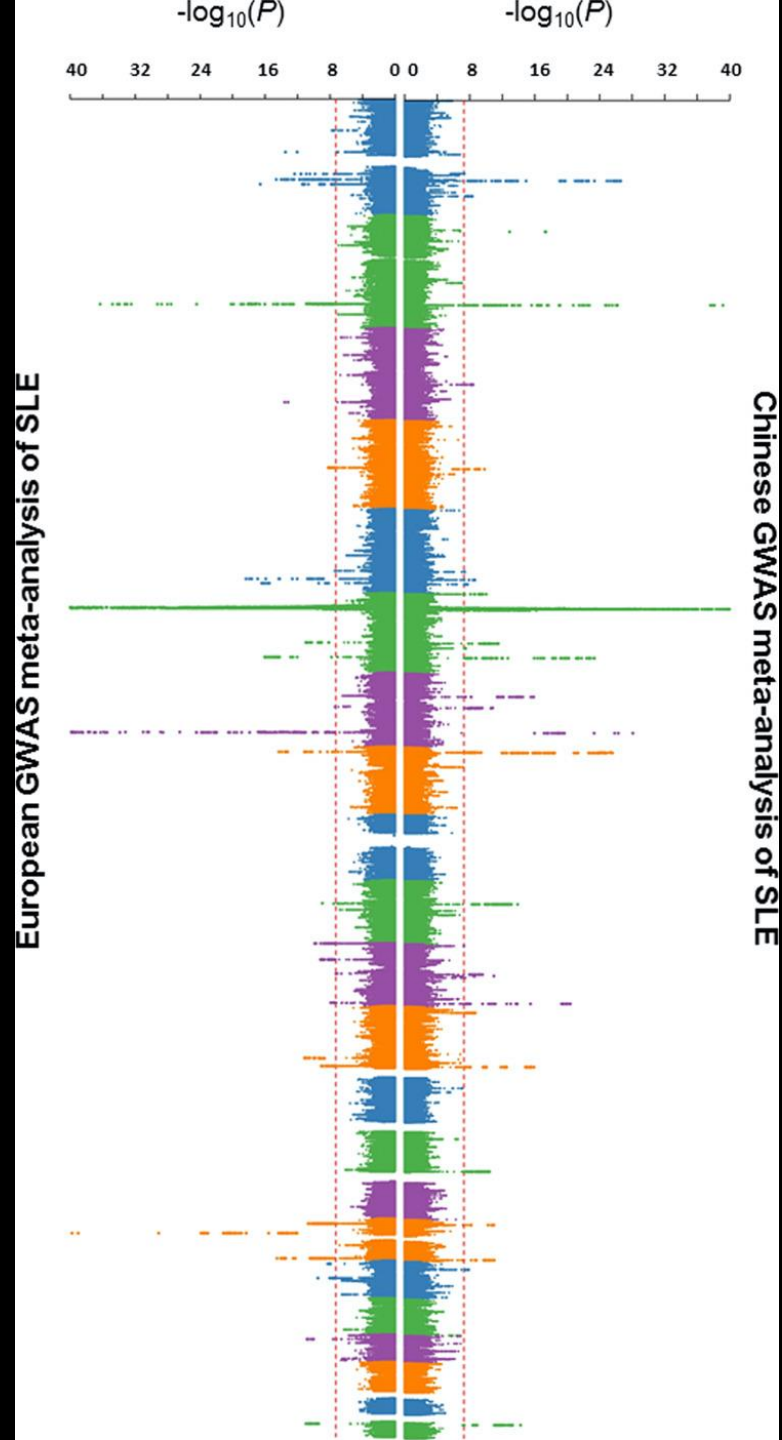
GWAS #1

follow-up approach:

“Focus on population differences”



GWAS in cohorts



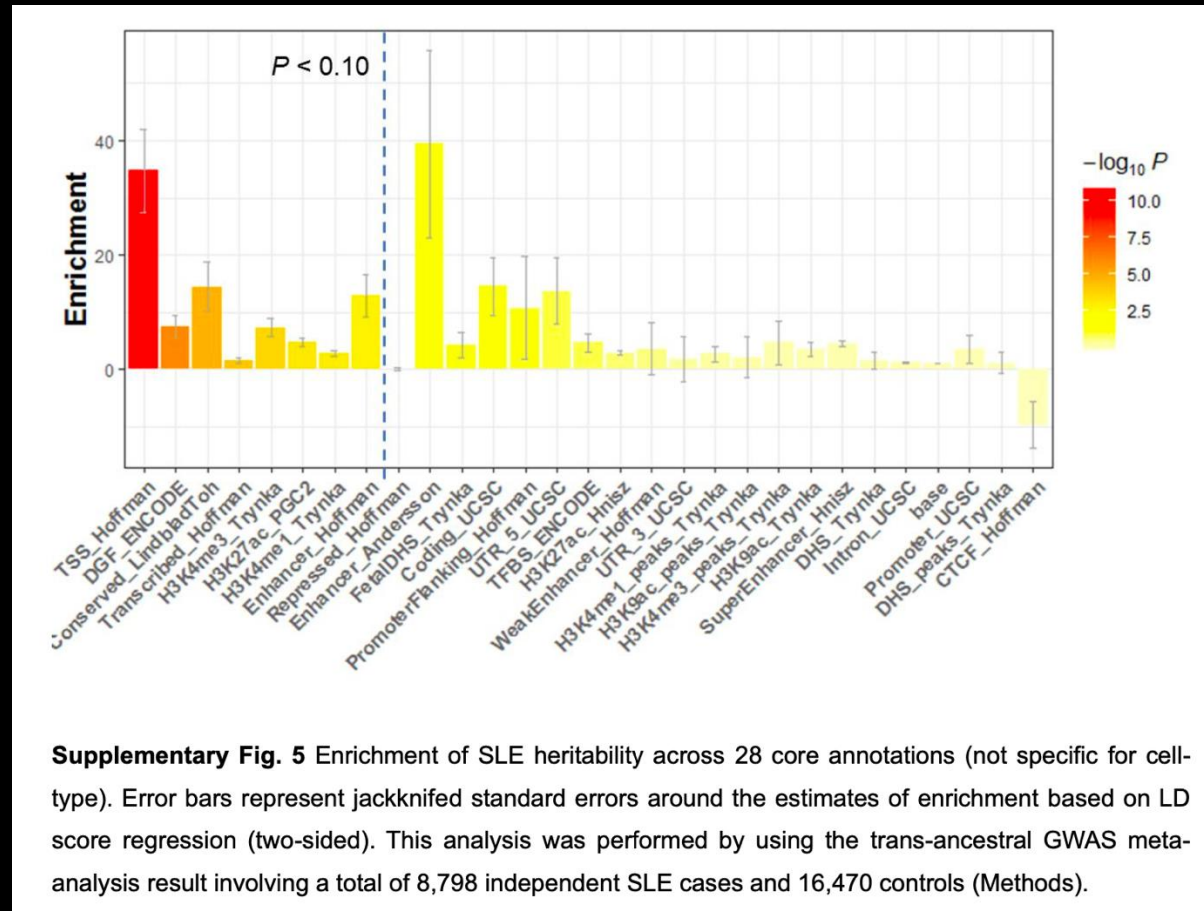
Meta-analysis in ancestral groups

Table 1

Summary association statistics of newly identified SLE-associated variants.

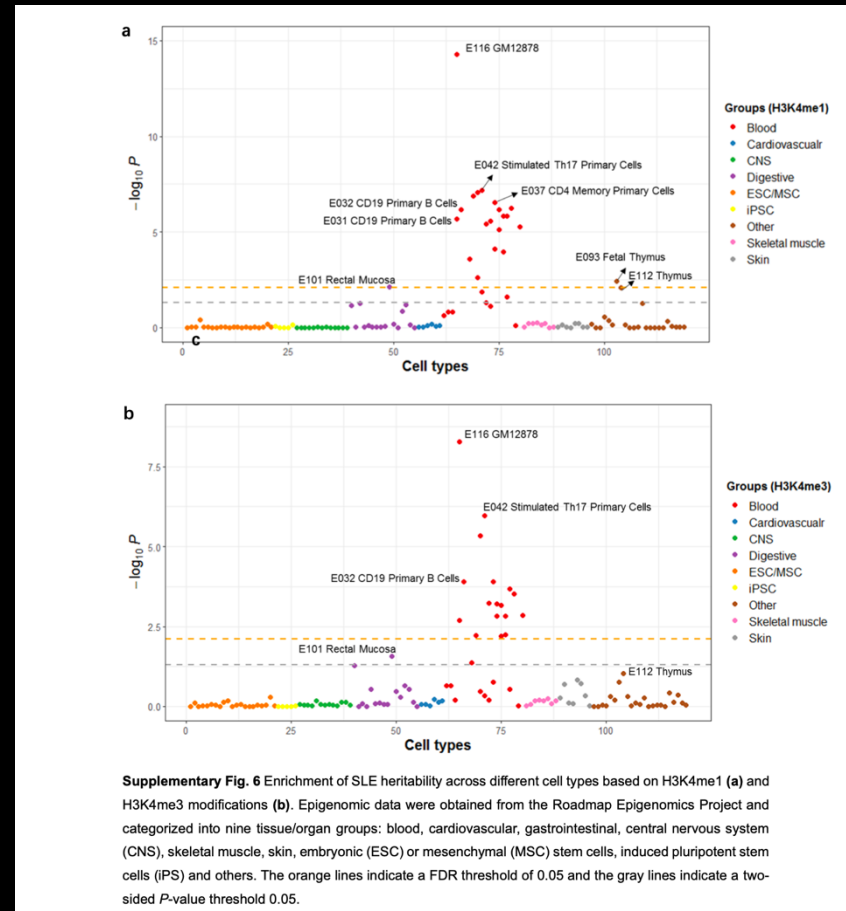
RSid	CHR	Pos (hg19)	AA ^a	AB ^a	Gene context	Annotation	EAS		EUR		Trans		Cochran's Q-test <i>P</i>
							OR ^b	<i>P</i>	OR ^b	<i>P</i>	OR ^b	<i>P</i>	
rs12093154	1	1178925	G	A	<i>CIQTNF12</i>	Missense	0.84	4.12E-06	0.82	1.38E-04	0.84	2.51E-09	0.648
rs3795310	1	8431607	C	T	<i>RERE</i>	Intronic	0.88	1.40E-03	0.88	8.03E-06	0.88	3.36E-08	0.92
rs28411034	1	38276997	G	A	<i>MTF1</i>	UTR3	0.86	5.80E-06	0.87	5.93E-06	0.86	1.50E-10	0.86
rs6702599	1	67825399	A	C	<i>IL12RB2</i>	Intronic	0.88	3.56E-02	0.82	1.78E-08	0.84	3.18E-09	0.302
rs1016140	1	117076547	G	T	<i>CD58</i>	Intronic	0.87	2.95E-10	1	9.62E-01	0.89	1.26E-08	0.007
rs11264750	1	157497160	A	G	<i>FCRL5</i>	Intronic	0.75	1.95E-12	0.81	1.81E-01	0.75	8.70E-13	0.655
rs12132445	1	170811799	G	A	<i>PRRX1, MROH9</i>	Intergenic	1.17	6.75E-08	1	9.67E-01	1.08	1.74E-04	1.05E-04
rs549669428	1	174895022	T	G	<i>RABGAP1L</i>	Intronic	0.75	1.85E-05	0.87	1.67E-04	0.84	4.53E-08	0.062
rs1547624	1	192543837	A	T	<i>RGS21, RGS1</i>	Intergenic	1.17	4.55E-08	1.06	1.38E-01	1.13	1.95E-07	0.025
rs2381401	2	144020974	C	T	<i>ARHGAP15</i>	Intronic	1.18	1.11E-07	1.11	1.97E-03	1.15	1.73E-09	0.221
rs9630991	2	191432139	G	A	<i>NEMP2, NAB1</i>	Intergenic	0.85	3.34E-06	0.85	6.73E-09	0.85	1.08E-13	0.971
rs11679484	2	198921604	C	A	<i>PLCL1</i>	Intronic	1.15	5.59E-05	1.1	1.91E-04	1.12	6.26E-08	0.413
rs3087243	2	204738919	G	A	<i>CTLA4</i>	Downstream	0.88	6.44E-06	0.91	6.61E-04	0.89	2.29E-08	0.396
rs438613	3	28072086	T	C	<i>LINC01980, CMC1</i>	Intergenic	1.13	6.46E-08	1.07	1.98E-02	1.1	1.32E-08	0.201
rs4690055	4	2748663	G	A	<i>TNIP2</i>	Intronic	0.91	4.14E-05	0.91	4.73E-04	0.91	5.28E-08	0.776
rs4697651	4	10721433	C	T	<i>CLNK, MIR572</i>	Intergenic	0.85	4.49E-05	0.9	3.26E-04	0.88	6.36E-08	0.256
rs6871748	5	35885982	T	C	<i>IL7R, CAPSL</i>	Intergenic	0.88	1.44E-05	0.9	6.40E-04	0.89	3.96E-08	0.579
rs6927090	6	252145	G	T	<i>DUSP22, IRF4</i>	Intergenic	1.23	3.48E-10	\	\	\	\	\
rs9387400	6	116694120	C	A	<i>DSE</i>	Intronic	0.74	3.14E-08	0.94	2.96E-02	0.89	5.89E-06	1.17E-04
rs13260060	8	71218360	G	A	<i>NCOA2</i>	Intronic	0.89	1.92E-08	0.99	7.55E-01	0.9	7.94E-08	0.044

Enrichment of heritability by annotation category (non cell-type-specific)



(LD score regression)

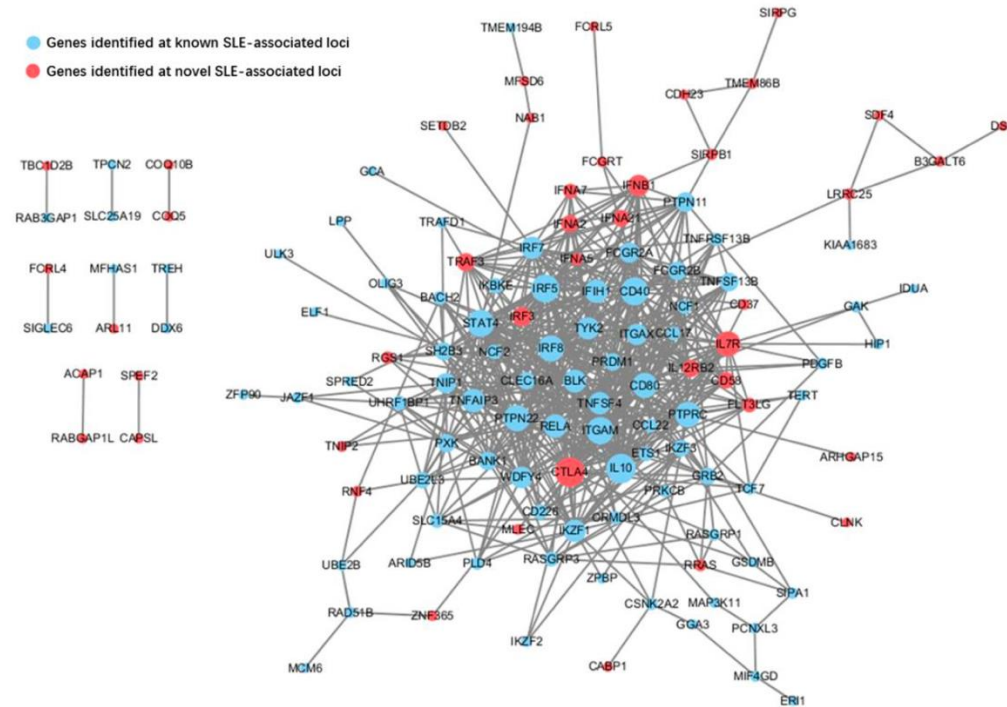
Enrichment of heritability by annotation category (cell-type specific)



(LD score regression)

Protein-interaction analysis

A significant level of protein-protein connectivity corresponding to genes found at the novel loci and known SLE-associated loci was observed ($P < 1E-16$; Supplementary Fig. 7; see “Methods” section)



Supplementary Fig. 7 Protein-Protein interaction (PPI) network corresponding to putative disease genes across SLE susceptibility loci. Genes identified at the novel SLE-associated loci are labeled in red, and genes found at the known SLE-associated loci are labeled in blue. The PPI information was retrieved from the STRING database (version 11.0).

Pathway (gene set) analysis

Forty-five pathways were significantly enriched with these putative SLE susceptibility genes (ToppGene²⁷, FDR < 0.05; Supplementary Table ³)

DEPICT was used to nominate putative causal genes.

Gene sets included KEGG, Reactome, and mouse knockout databases.

Supplementary Table 3 Pathway enrichment analysis based on putative SLE genes

Name	Source	P-value	FDR
Cytokine Signaling in Immune system	REACTOME	3.05E-12	2.18E-08
Interferon alpha/beta signaling	REACTOME	2.23E-11	7.98E-08
Toll-like receptor signaling pathway	KEGG	1.34E-10	2.91E-07
Measles	KEGG	1.98E-10	2.91E-07
RIG-I/MDA5 mediated induction of IFN-alpha/beta pathways	REACTOME	2.03E-10	2.91E-07
RIG-I-like receptor signaling pathway	KEGG	5.83E-10	6.54E-07
TRAF6 mediated IRF7 activation	REACTOME	6.41E-10	6.54E-07
Regulation of IFNA signaling	REACTOME	2.25E-09	1.80E-06
Cytokine-cytokine receptor interaction	KEGG	2.27E-09	1.80E-06
Hepatitis B	KEGG	5.93E-09	4.24E-06
Herpes simplex infection	KEGG	1.06E-08	6.91E-06
Jak-STAT signaling pathway	KEGG	1.47E-08	8.76E-06
Autoimmune thyroid disease	KEGG	2.11E-08	1.16E-05
Hepatitis C	KEGG	2.44E-08	1.25E-05
Cytosolic DNA-sensing pathway	KEGG	9.71E-08	4.63E-05
TRAF3-dependent IRF activation pathway	REACTOME	2.17E-07	9.71E-05
Innate Immune System	REACTOME	3.08E-07	1.30E-04
NOD-like receptor signaling pathway	KEGG	3.55E-07	1.41E-04
Influenza A	KEGG	4.23E-07	1.59E-04
Interferon Signaling	REACTOME	1.96E-06	6.68E-04
Signal regulatory protein (SIRP) family interactions	REACTOME	4.17E-06	1.35E-03
Factors involved in megakaryocyte development and platelet production	REACTOME	1.16E-05	3.33E-03
Adaptive Immune System	REACTOME	1.25E-05	3.45E-03
Activation of IRF3/IRF7 mediated by TBK1/IKK epsilon	REACTOME	1.72E-05	4.40E-03
Ovarian tumor domain proteases	REACTOME	1.87E-05	4.60E-03
Signaling by Interleukins	REACTOME	2.05E-05	4.88E-03
Osteoclast differentiation	KEGG	2.18E-05	5.03E-03
NF-kappa B signaling pathway	KEGG	2.27E-05	5.08E-03
Natural killer cell mediated cytotoxicity	KEGG	2.57E-05	5.57E-03
Toll Like Receptor 3 (TLR3) Cascade	REACTOME	2.98E-05	5.91E-03
MyD88-independent TLR3/TLR4 cascade	REACTOME	2.98E-05	5.91E-03
TRIF-mediated TLR3/TLR4 signaling	REACTOME	2.98E-05	5.91E-03
Intestinal immune network for IgA production	KEGG	7.49E-05	1.28E-02
Activated TLR4 signalling	REACTOME	7.78E-05	1.29E-02
Epstein-Barr virus infection	KEGG	8.80E-05	1.43E-02
Toll Like Receptor 4 (TLR4) Cascade	REACTOME	1.38E-04	2.10E-02
Hemostasis	REACTOME	1.87E-04	2.57E-02
Ras signaling pathway	KEGG	2.05E-04	2.75E-02
Tuberculosis	KEGG	2.08E-04	2.75E-02
Signaling by the B Cell Receptor (BCR)	REACTOME	2.74E-04	3.56E-02

Negative regulators of RIG-IMDA5 signaling	REACTOME	2.93E-04	3.74E-02
Primary immunodeficiency	KEGG	3.26E-04	3.89E-02
T cell receptor signaling pathway	KEGG	3.26E-04	3.89E-02
Downstream signaling events of B Cell Receptor (BCR)	REACTOME	4.26E-04	4.99E-02
B cell receptor signaling pathway	KEGG	4.37E-04	4.99E-02

The statistics are provided by ToppGene (<https://toppgene.cchmc.org/>). P-value is two-sided. FDR is the P-value after adjusting multiple testing.

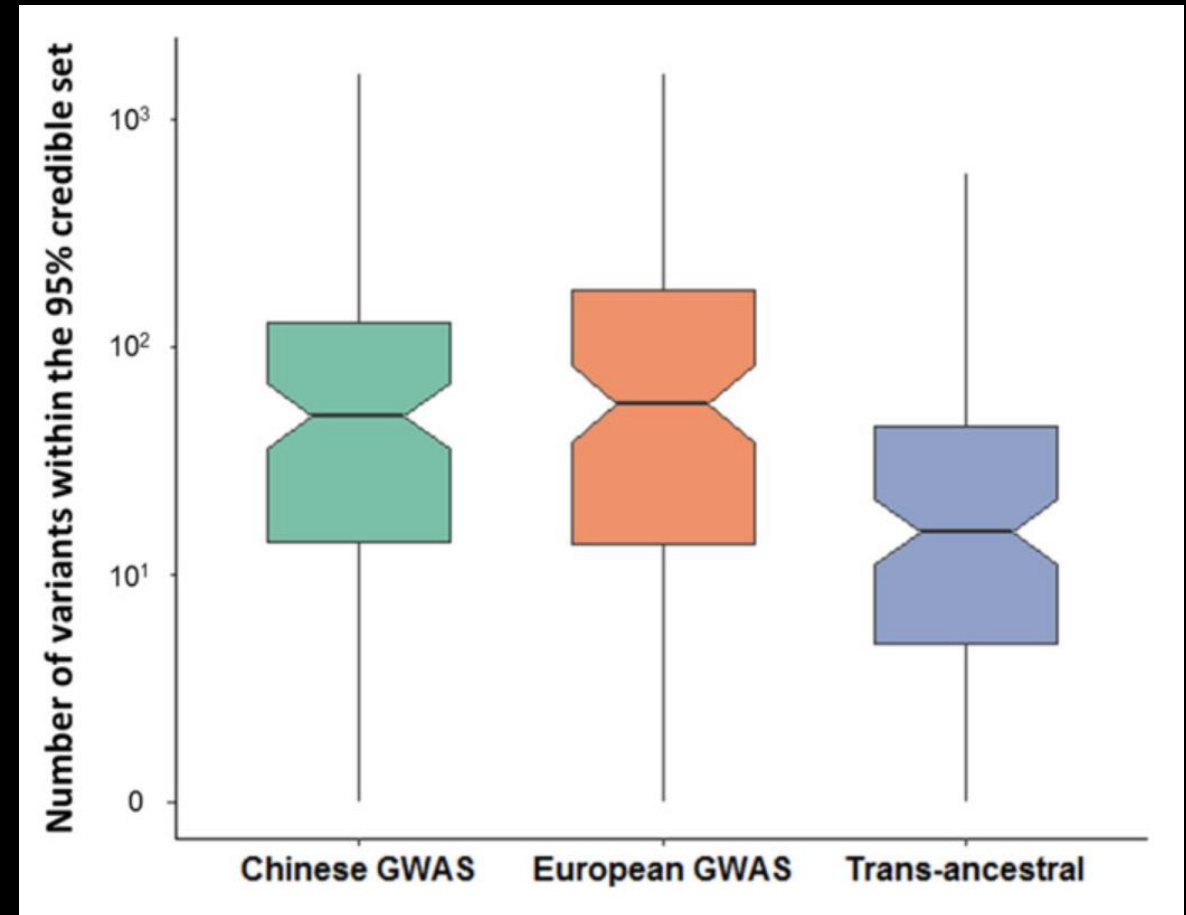
Trans-ancestral fine-mapping

Fine-mapping across 108 SLE-associated loci based on the association results from the Chinese SLE GWAS, the European GWAS, and the trans-ancestral meta-analyses. The Y-axis indicates the number of potential causal variants at each locus based on the 95% credible sets of the association results

(Fine-mapping done with PAINTOR)

Takeaway: trans-ethnic meta-analysis results appear to help with prioritization (smaller credible set when combining LD across two populations)

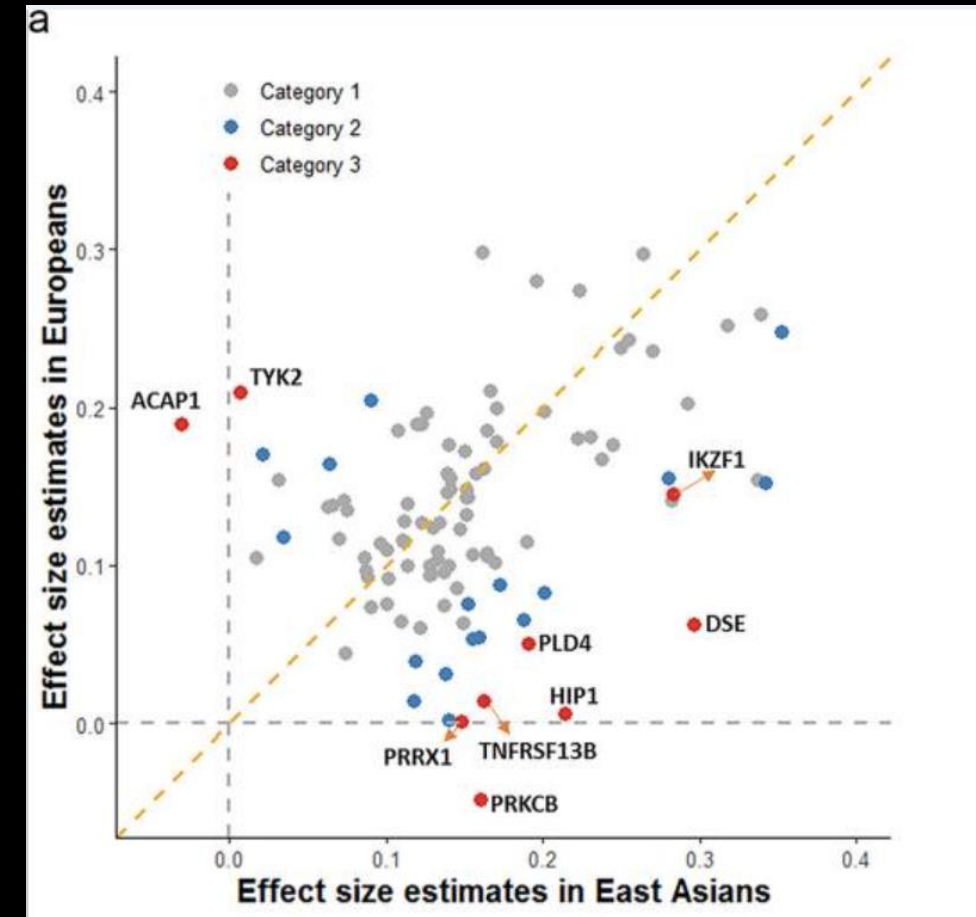
“A single putative causal variant was identified for the *WDFY4* and *TNFSF4* loci, the latter of which was functionally validated in a previous study”



Exploration of heterogeneity

Based on the analysis of Cochran's Q (CQ)-test that assesses heterogeneity of effect-size estimates from different ancestral groups, SLE-associated variants or loci outside of the HLA region were divided into four categories:

- (1) **ancestry-shared disease loci** tagged by variants with CQ-test $P \geq 0.05$;
- (2) **putative ancestry-heterogeneous disease loci** with CQ-tests of $P < 0.05$ but FDR adjusted CQ-test $P \geq 0.05$;
- (3) **ancestry-heterogeneous disease loci** with FDR adjusted CQ-test $P < 0.05$; and
- (4) **disease loci tagged by associated variants with the risk allele absent** in one of the two ancestries



Only loci with MAF > 1% in both populations included

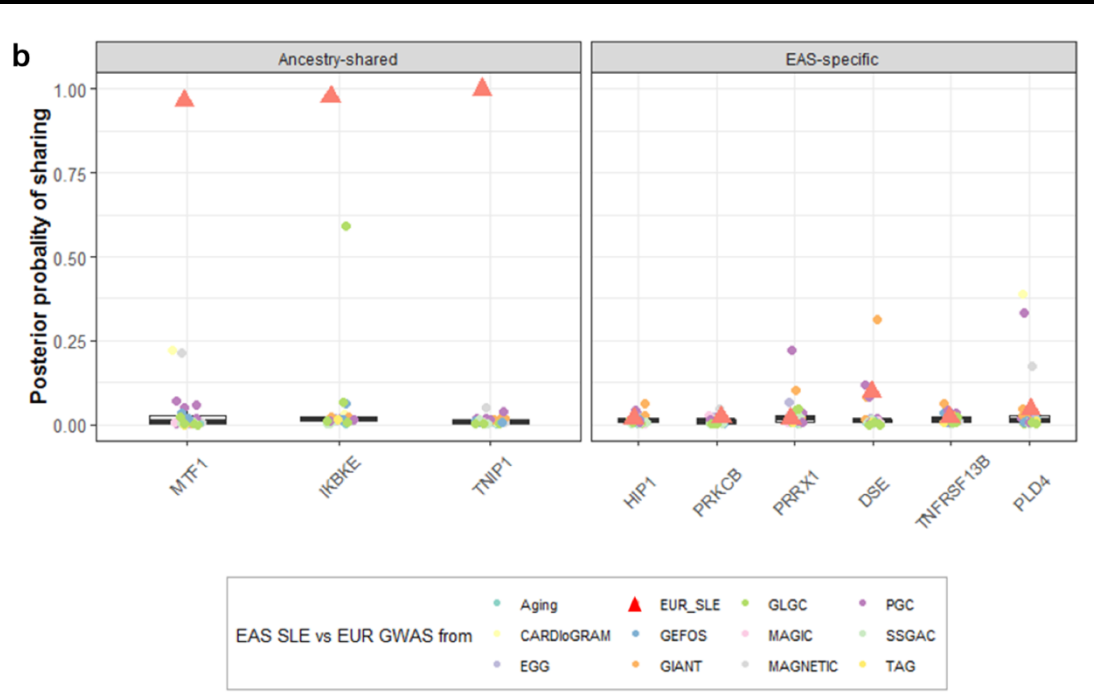
“The ancestry-heterogeneous loci appear to be enriched for functions related to antibody production.”

Literature search for the 9 ancestry-heterogeneous loci (red dots) in human/mouse

Exploration of heterogeneity

COLOC was used to study colocalization of the European and Chinese associations.

Colocalization methods consider many SNPs, rather than only the leading variant in a locus, to compare association signals between ancestral groups. The ancestry-shared disease loci showed much higher posterior probabilities (PP) of colocalization (mean of PP = 0.42) than the ancestry-specific loci.



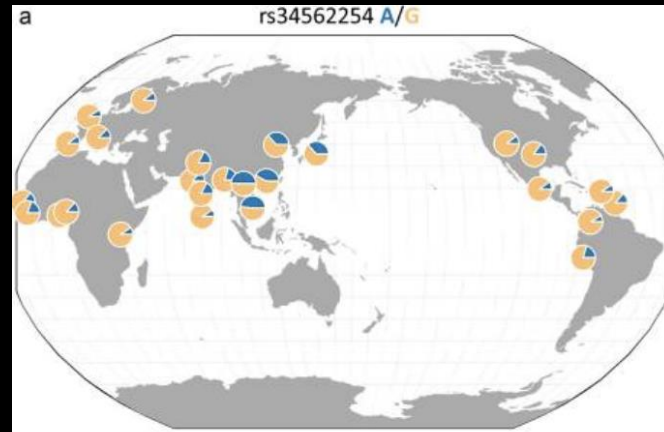
Supplementary Fig. 11 b,

Red triangles represent the posterior probability of SLE association signals between East Asian and European ancestries. Other symbols represent the colocalization of Chinese SLE association signals with those in European populations for 27 phenotypes unrelated to immunity.

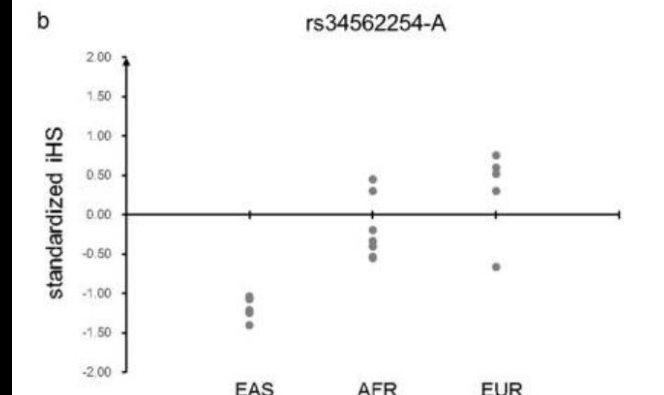
Exploration of heterogeneity

Risk allele frequency and standardized integrated haplotype scores (iHS) across different populations for the Asian-specific variant at *TNFRSF13B* locus [iHS = measure of positive selection].

a Frequency for the risk allele rs34562254-A across populations.



b Standardized iHS for the variants across different continental populations (EAS: East Asians; AFR: Africans; EUR: Europeans). The risk allele rs34562254-A is a derived allele, and negative iHS value indicates that the haplotypes carrying the derived allele are longer than the haplotypes carrying the ancestral allele. The frequency and iHS were calculated using data from the 1000 Genomes Project.



→ Suggests recent positive selection for the risk allele in East Asians.

BAFF (*TNFSF13B*) and its receptors, one of which is TACI (*TNFRSF13B*), play essential roles in B cell survival and differentiation

“Adaptation of the host to resist pathogens may underlie some of the ancestral group-heterogeneity”

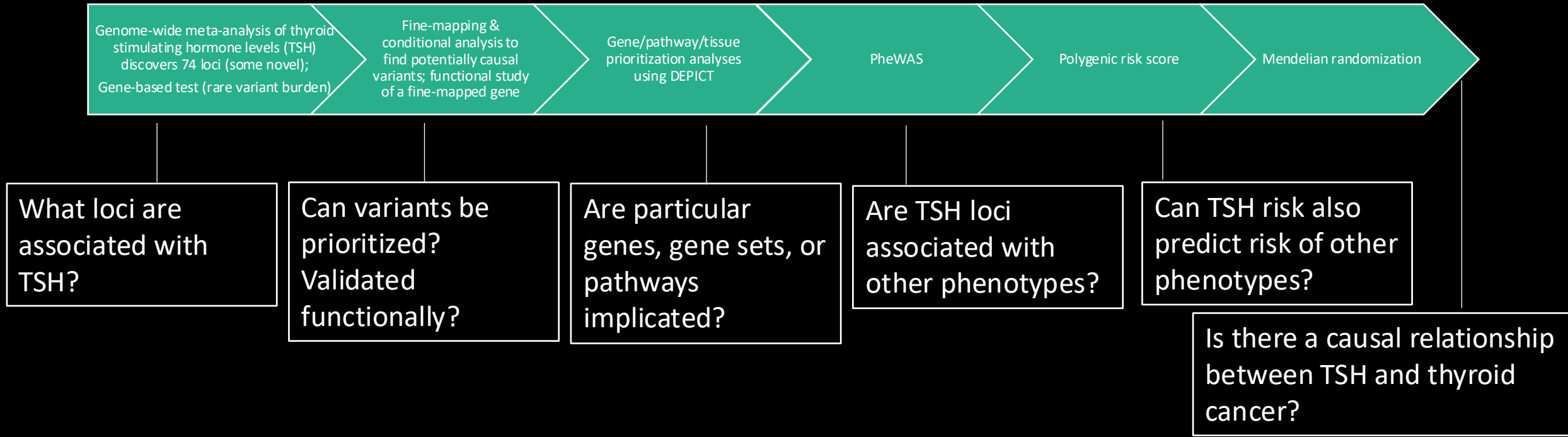
GWAS #2

follow-up approach:

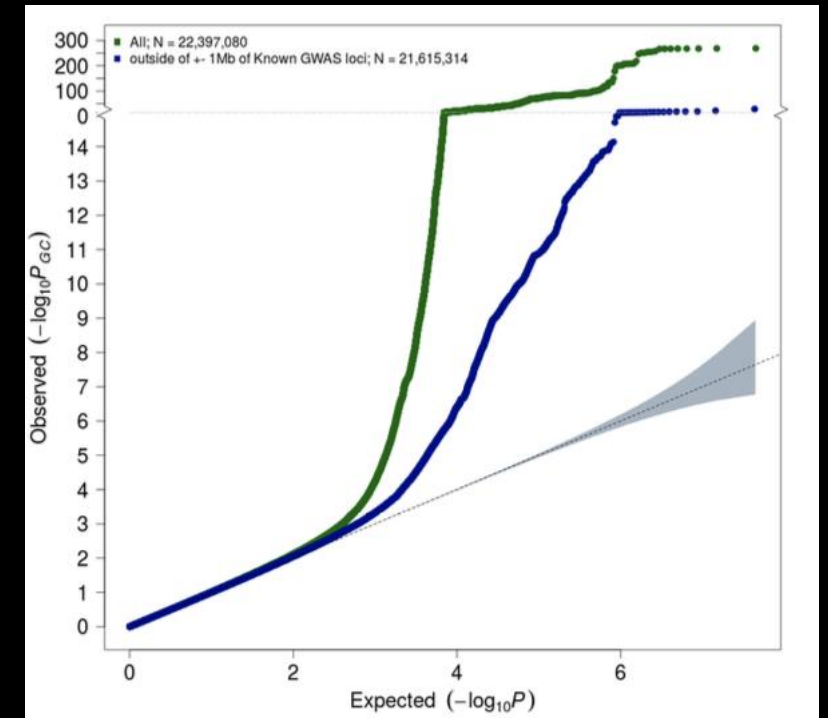
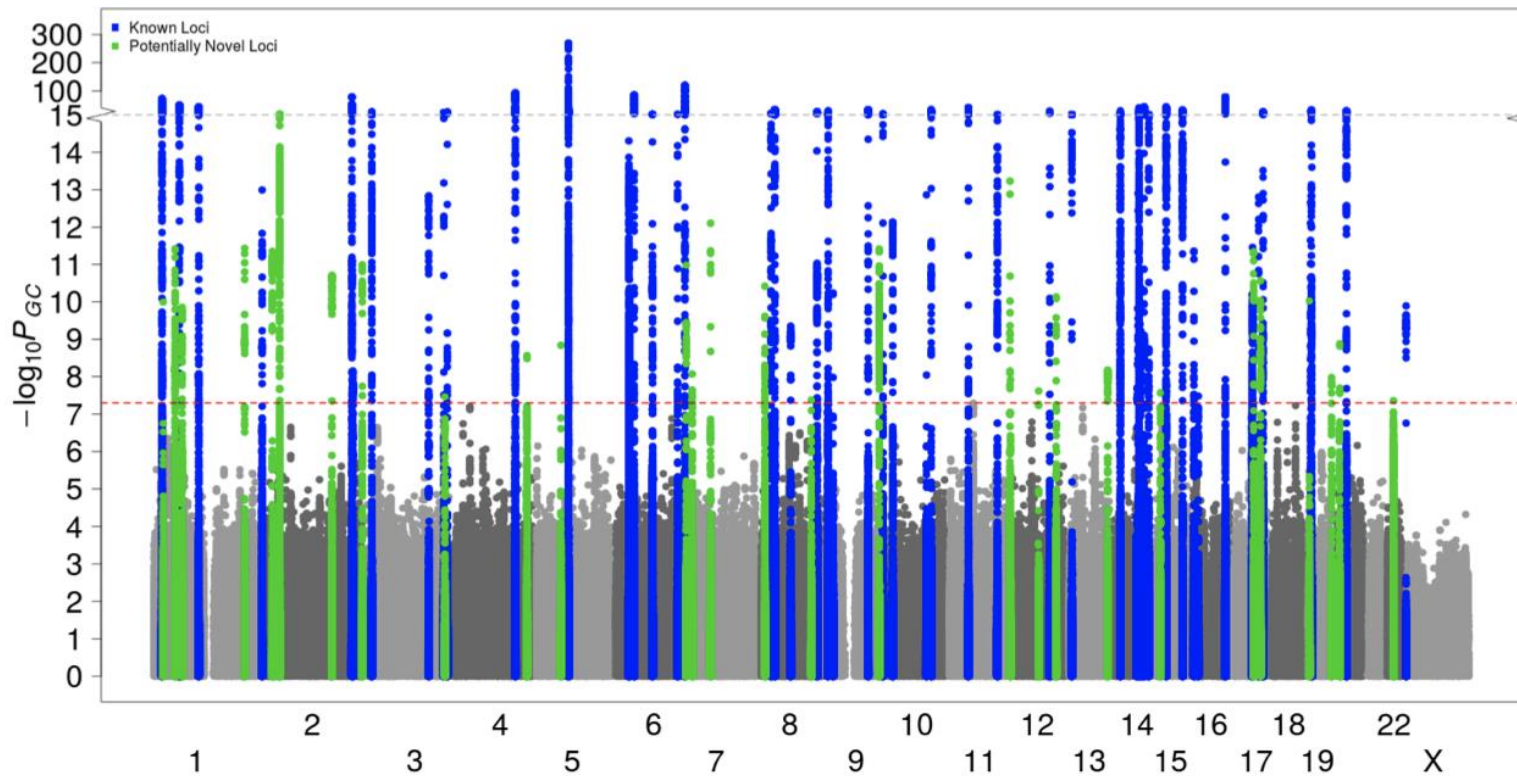
focus on shared genetics with another trait

Zhou W, Brumpton B, Kabil O, et al. GWAS of thyroid stimulating hormone highlights pleiotropic effects and inverse association with thyroid cancer. *Nat Commun.* 2020;11(1):3981. Published 2020 Aug 7.
doi:10.1038/s41467-020-17718-z

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7414135/>

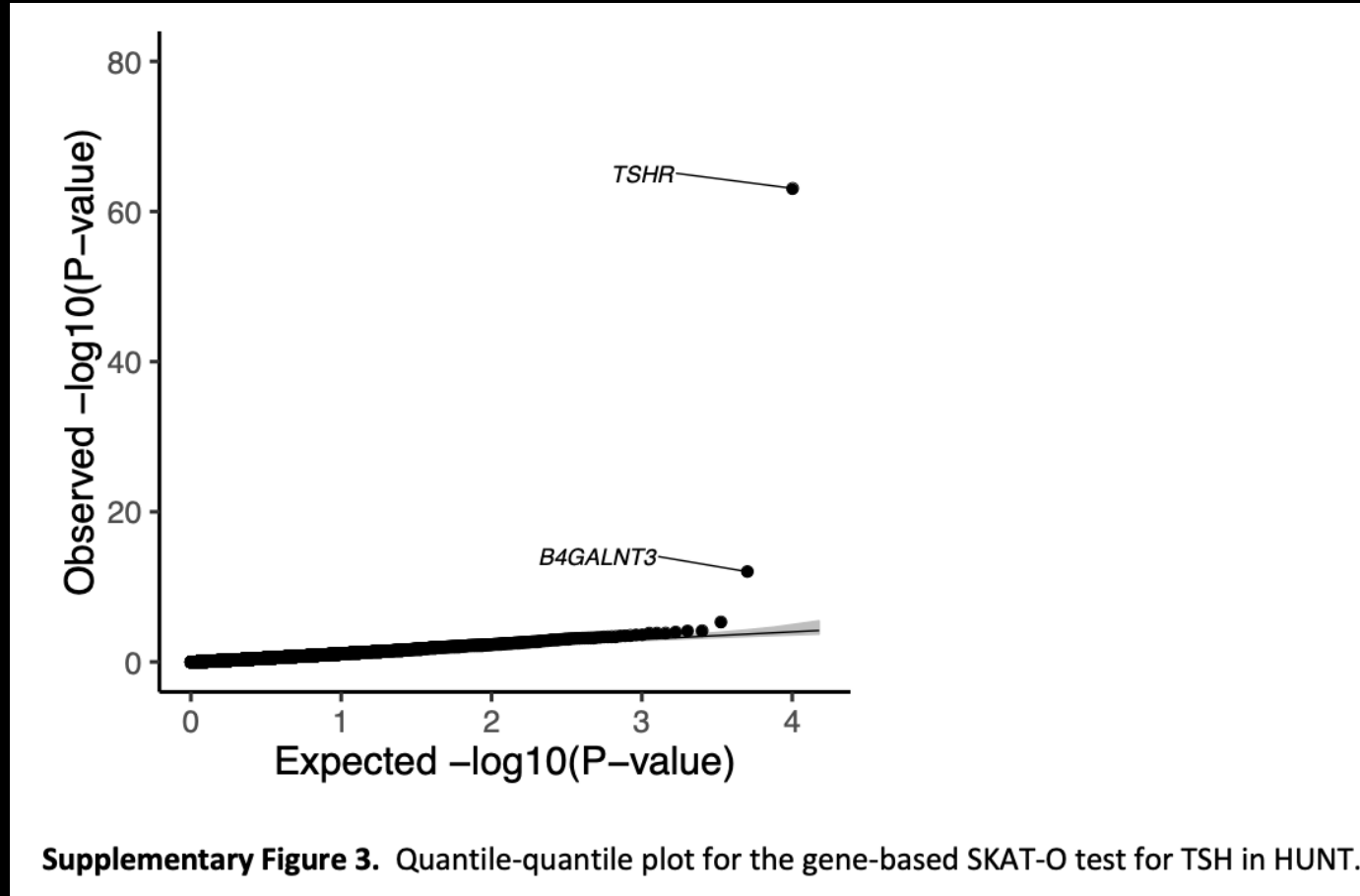


GWAS of TSH



Supplementary Figure 1. Manhattan and Q-Q Plots for the meta-analysis GWAS of TSH from the HUNT study (N = 55,342), the MGI study (N = 10,085), and the ThyroidOmics consortium (N = up to 54,288). N: sample size.

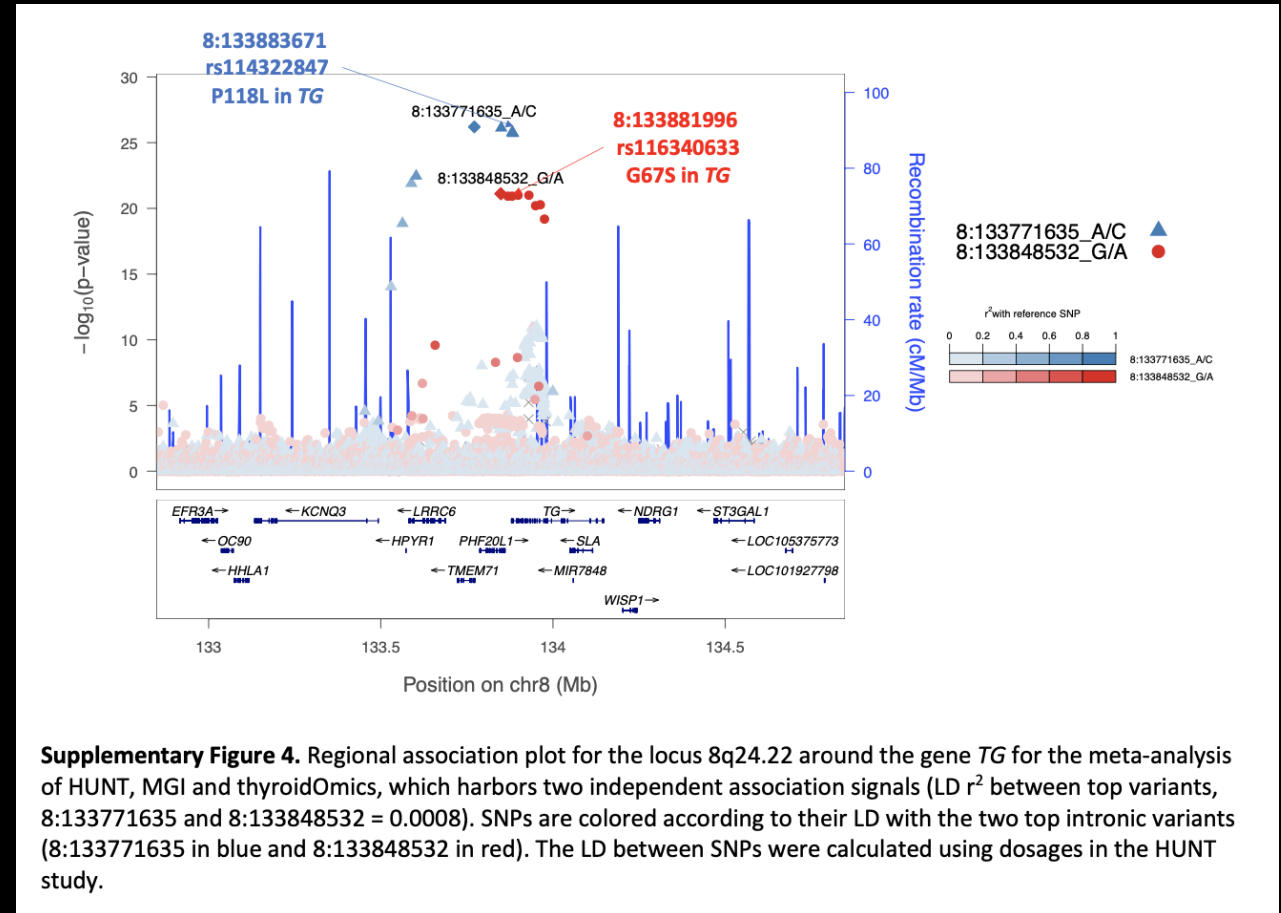
Gene-based burden test



Fine-mapping an interesting locus (functional follow-up not shown here)

To identify potentially causal variants at TSH loci, we conducted fine-mapping using SuSiE⁴⁰, which estimates the number of causal variants and obtains credible sets of variants with 95% cumulative posterior probability through Iterative Bayesian Stepwise Selection...

...identified two association signals in the locus of the thyroglobulin gene (*TG*)



Prioritization/gene set analyses

Input: 161 GWAS loci with $p < 1E-5$ (as LD blocks)

Tissue associations:

- None significant (Bonferroni or FDR)
- Most significant: membranes, thyroid gland (as expected)

Gene prioritization:

- “Based on functional similarity to other genes among TSH loci, 70 genes at the TSH-associated loci were prioritized by DEPICT with $FDR \leq 0.01$ ” (including several novel associations)

Gene set analysis

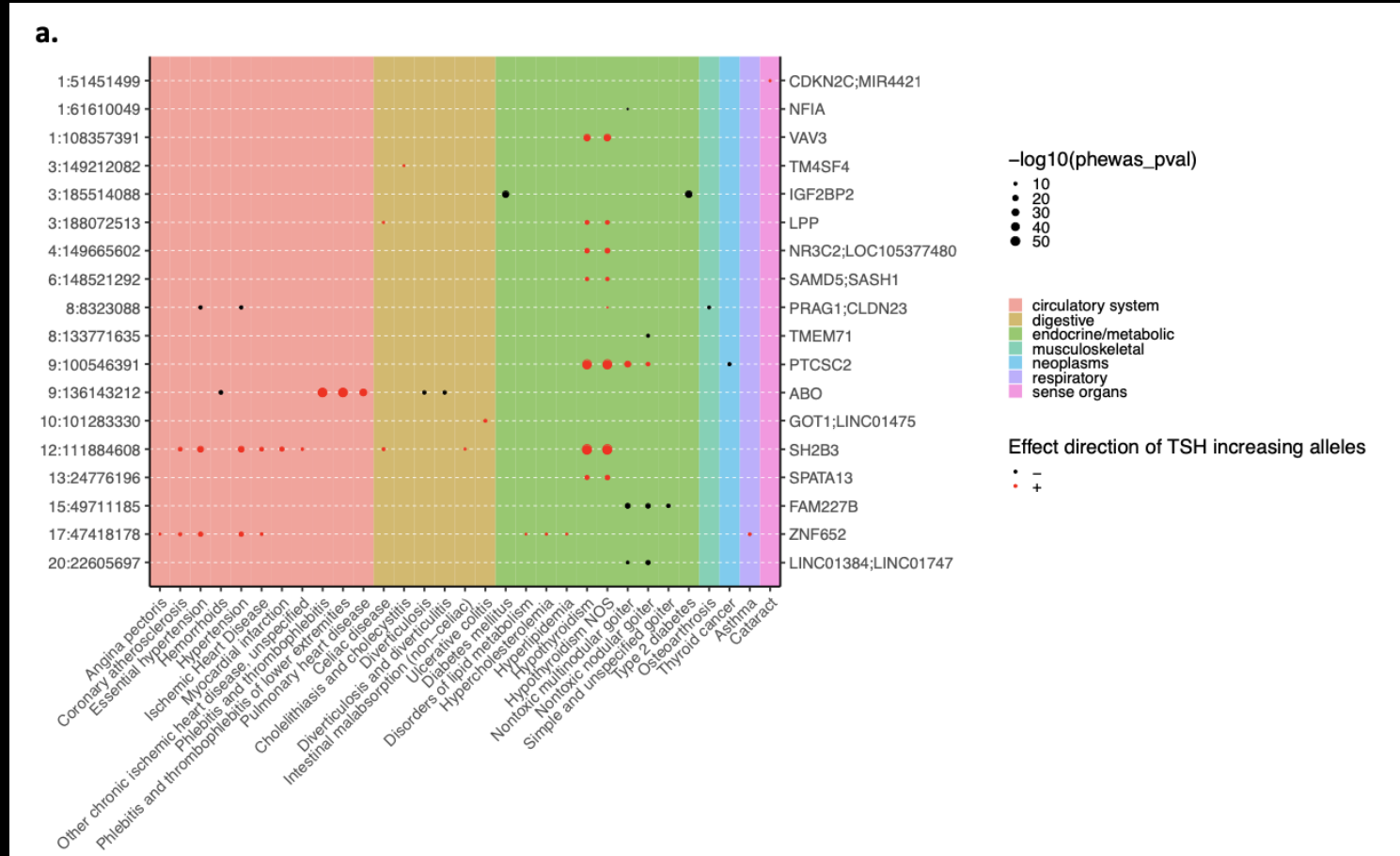
- “CTSD PPI subnetwork”
- “regulation of phosphorylation”

PheWAS

- 95 top non-HLA variants
- Test for association with 1283 diseases (phecodes) and 274 continuous traits in UKBB

PheWAS

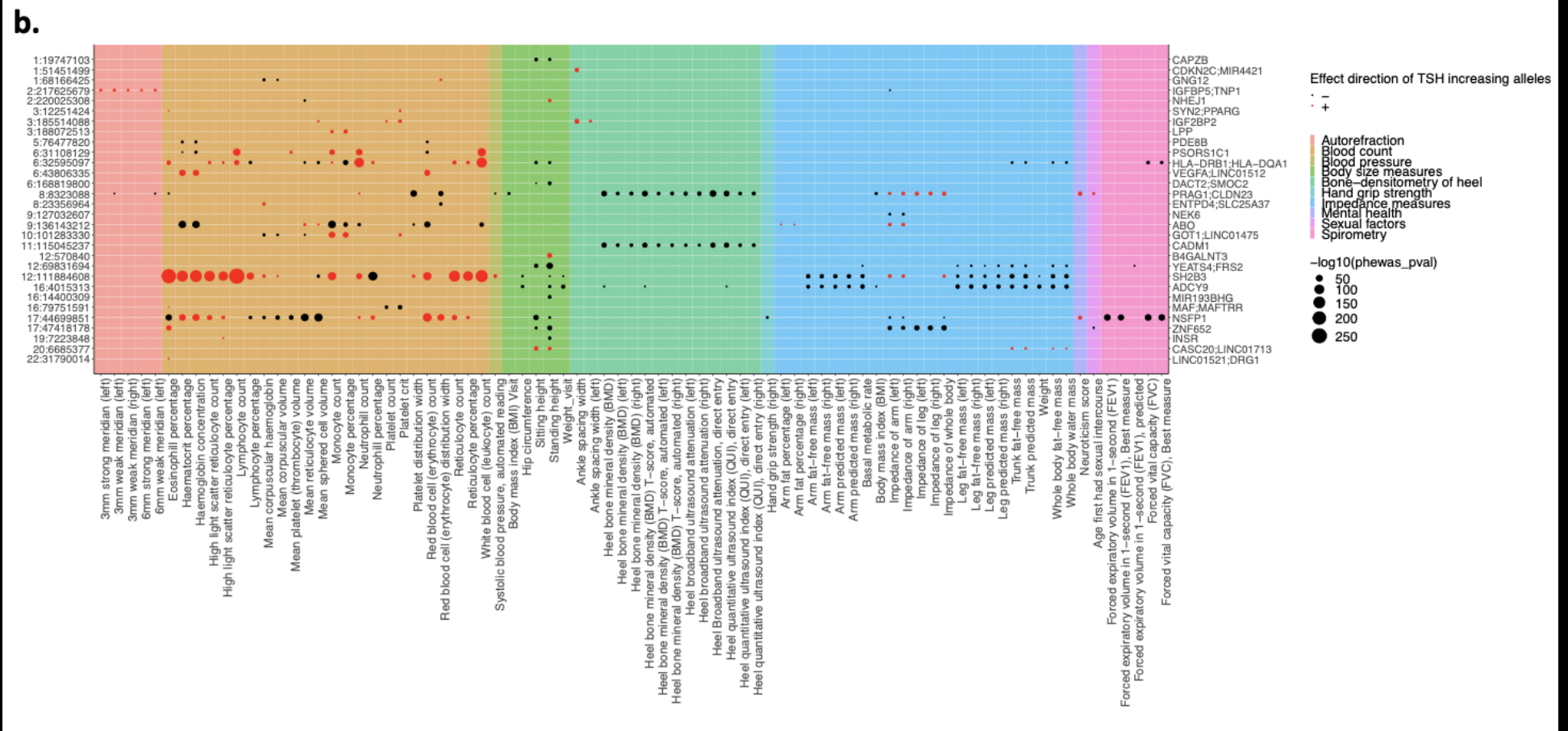
We identified significant (P -value $< 5 \times 10^{-8}$) pleiotropic association for 18 out of 95 non-HLA variants across 31 disease phenotypes including thyroid disorders, diabetes, cardiovascular disease, digestive system disorders, asthma, and cataracts



PheWAS

34 non-HLA variants were significantly associated (P -value $< 5 \times 10^{-8}$) with one or more quantitative traits, including body mass index, lung function measurements, metrics of bone density, spherical power/meridian measurements, and blood cell counts

TSH-increasing alleles at one or more loci were associated with an increased risk of cardiovascular disease, smaller body size, reduced bone mineral density, decreased lung function, and an increased risk of hypothyroidism and a decreased risk of goiter. These results are generally consistent with previous studies



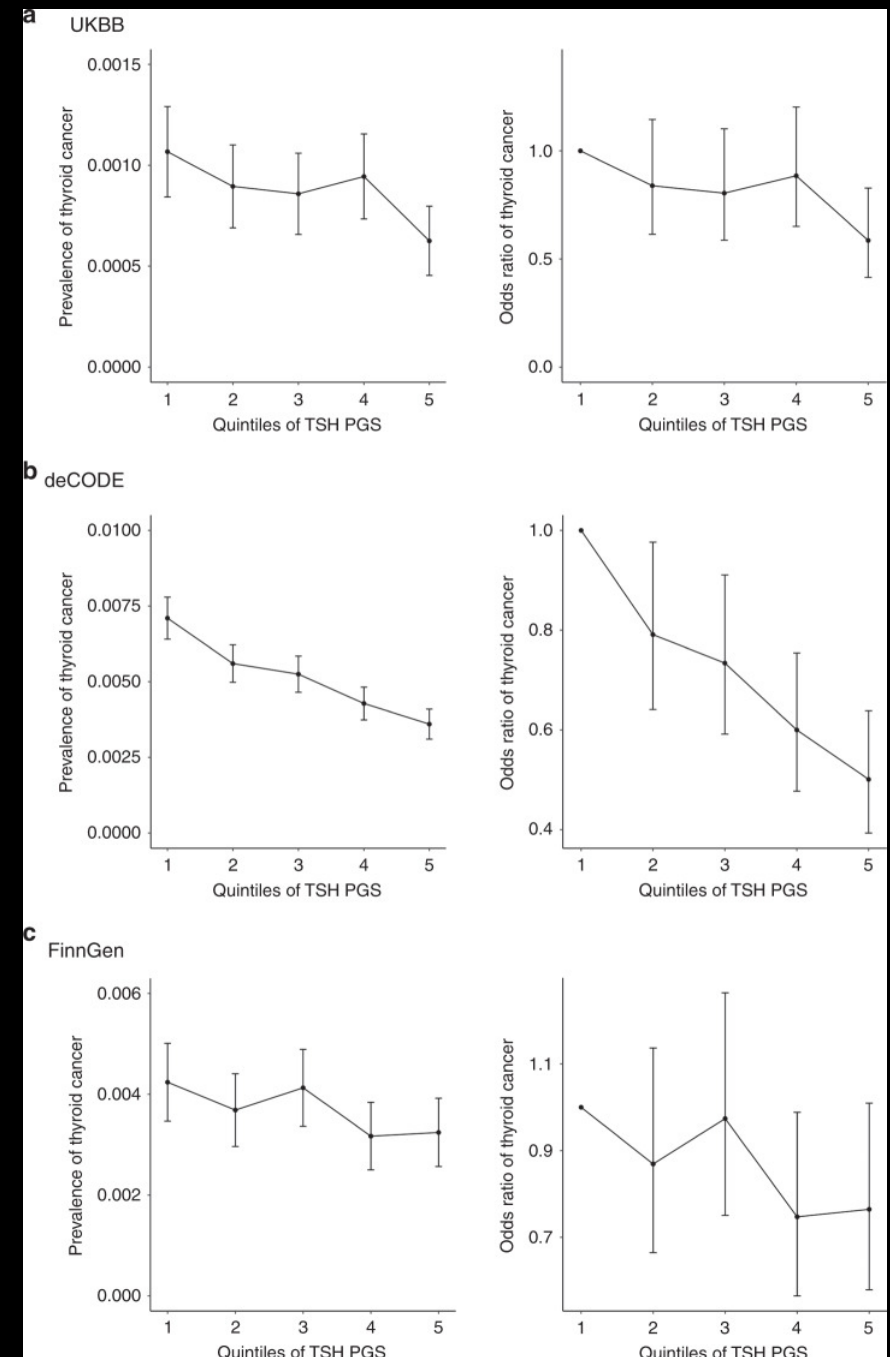
PRS PheWAS

“Although individual TSH variants may exhibit pleiotropic effects, it is also possible that the cumulative effects of TSH-modifying genetic variants may lead to disease. Therefore, we constructed PGS from the 95 independent ... and examined their association with the 1283 human diseases constructed from International Classification of Diseases (ICD) codes in the UKBB”

PRS PheWAS

The risk of thyroid cancer is lower for individuals with genetically predicted higher TSH levels.

Plots of thyroid cancer prevalence by quintiles of TSH PGS (left) and odds ratio of thyroid cancer in relation to the lowest quintile (right) in data sets UKBB (a; N case = 358, N control = 407,399), deCODE (b; N case = 1003, N control = 278,991)¹⁹, and FinnGen (c; N case = 501, N control = 135,137). N : sample size. N case: sample size of cases. N control: sample size of controls. Error bars represent 95% confidence intervals.



Mendelian randomization

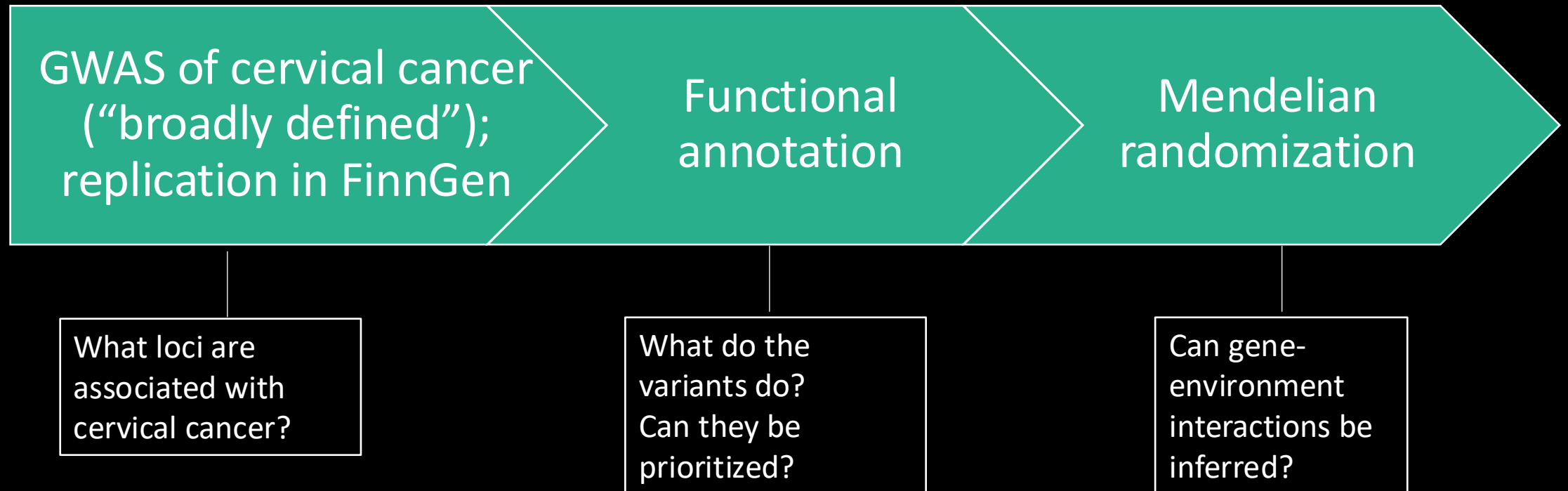
“Two-sample Mendelian randomization using TSH index variants as instrumental variables suggests a protective effect of higher TSH levels (indicating lower thyroid function) on risk of thyroid cancer and goiter. Our findings highlight the pleiotropic effects of TSH-associated variants on thyroid function and growth of malignant and benign thyroid tumors.”

GWAS #3

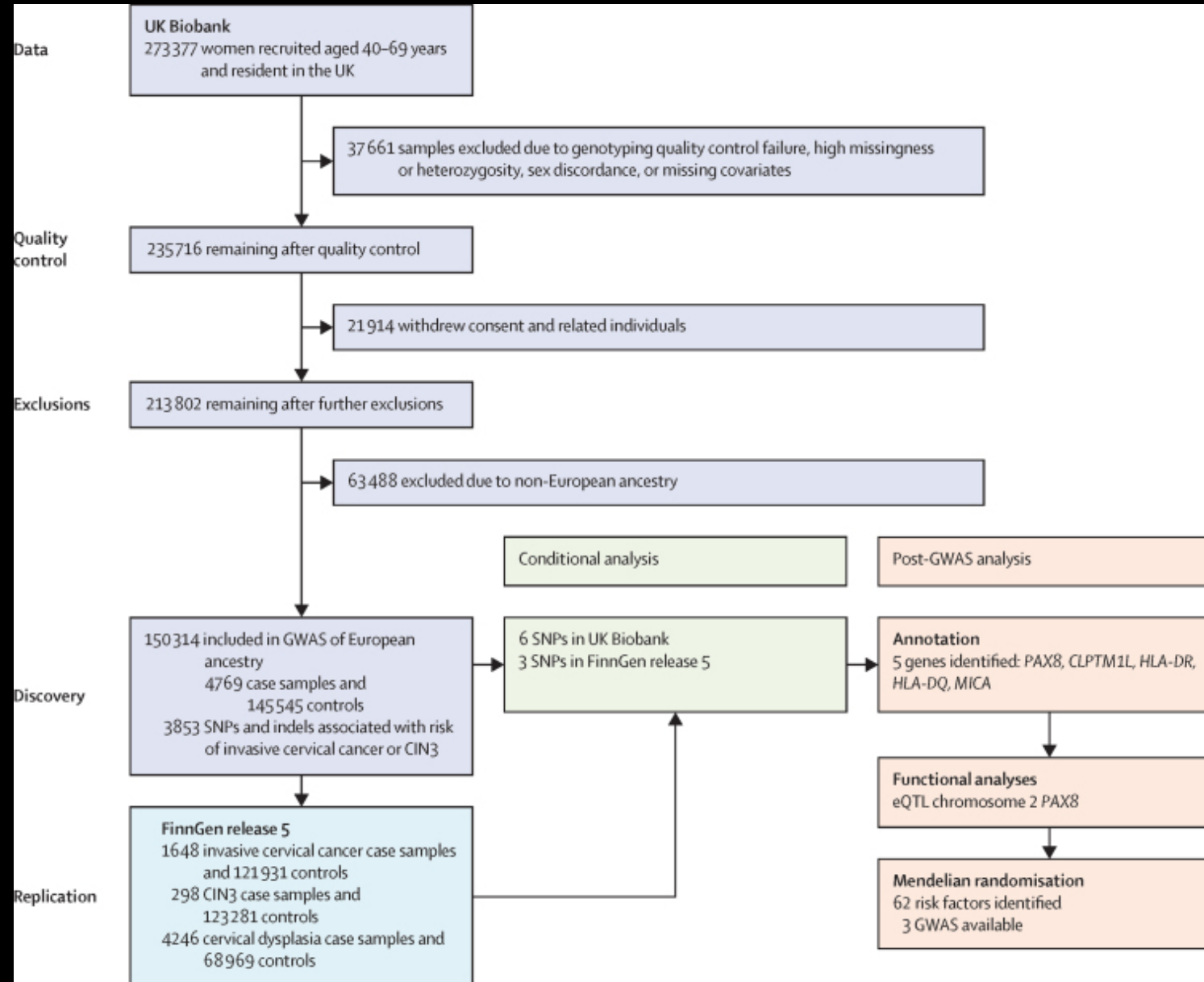
Follow-up approach:

A little annotation and MR

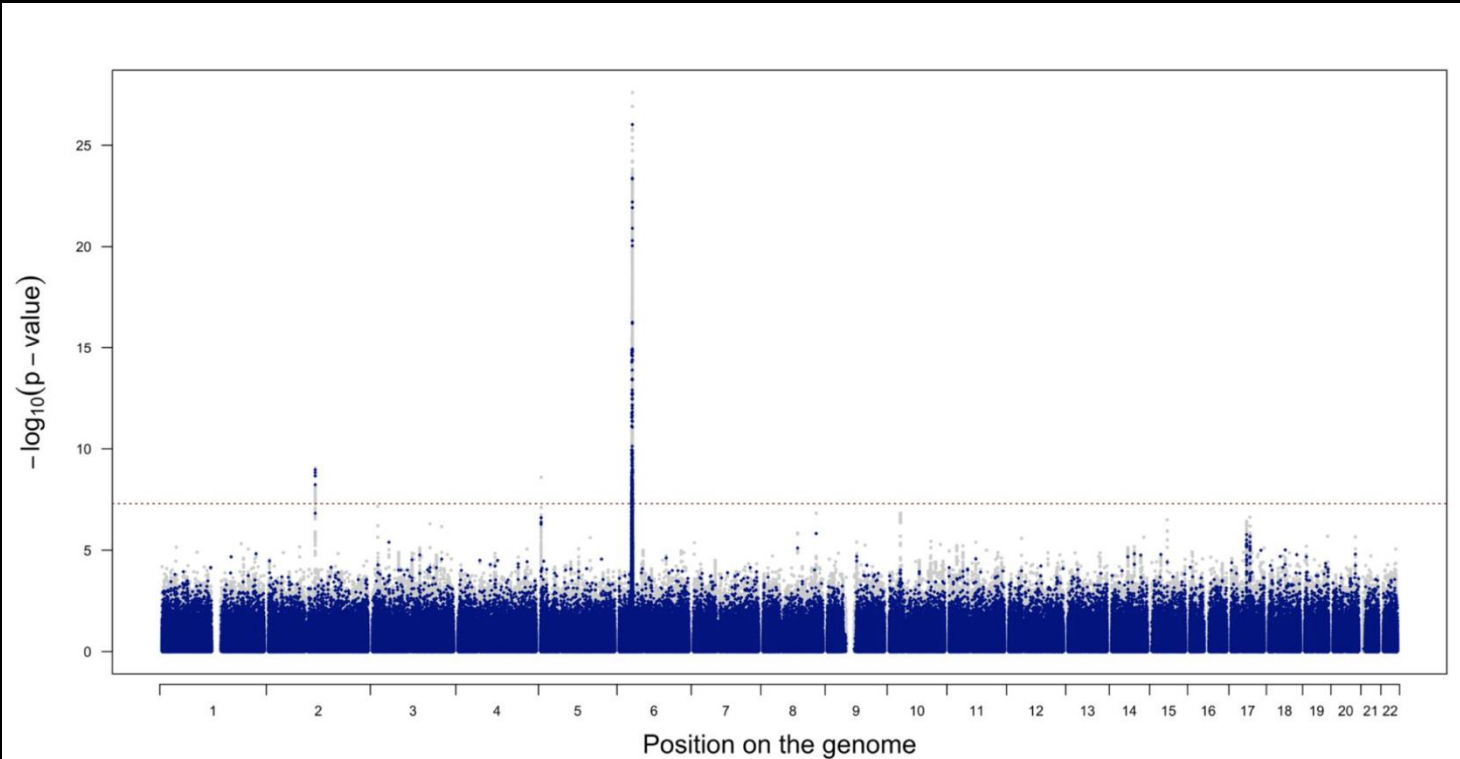
Bowden, S. J., Bodinier, B., Kalliala, I., Zuber, V., Vuckovic, D., Doulgeraki, T., Whitaker, M. D., Wielscher, M., Cartwright, R., Tsilidis, K. K., Bennett, P., Jarvelin, M. R., Flanagan, J. M., Chadeau-Hyam, M., Kyrgiou, M., & FinnGen consortium (2021). Genetic variation in cervical preinvasive and invasive disease: a genome-wide association study. *The Lancet. Oncology*, 22(4), 548–557. [https://doi.org/10.1016/S1470-2045\(21\)00028-0](https://doi.org/10.1016/S1470-2045(21)00028-0)



GWAS

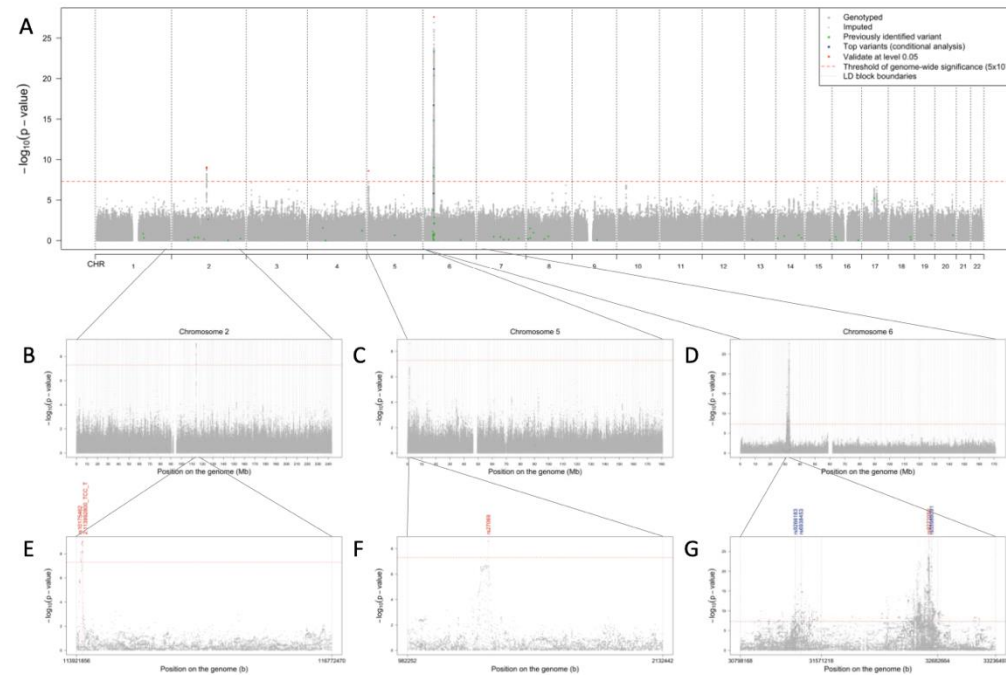


GWAS



Supplementary Figure 5. Manhattan plots showing the associations between genetic variants and cervical cancer status in the European unrelated population (N=4,797 cases, of which 764 ICC, vs N=146,178 controls). Genetic variants are ordered by their position on the genome (X-axis). The p-values, measuring the strength of association, are derived from additive logistic models and are represented on the $-\log_{10}$ scale (Y-axis). The genome-wide significance level is set to 5×10^{-8} (horizontal red line). Results are presented for all the (N=9,600,464) genotyped and imputed variants (grey), and the (N=656,284) genotyped only variants (blue).

GWAS



Supplementary Figure 1. Manhattan plots showing the associations between genetic variants and CIN3/ICC in the European unrelated population (N=4,797 cases vs N=146,178 controls). Genetic variants are ordered by their position on the genome (X-axis). The p-values, measuring the strength of association, are derived from additive logistic models and are represented on the $-\log_{10}$ scale (Y-axis). The genome-wide significance level is set to 5×10^{-8} (horizontal red line). Results are presented for all the (N=9,600,464) assayed and imputed variants (A), for chromosome 2 (N=798,812 variants, B), 5 (N=616,800 variants, C), and 6 (N=651,104 variants, D). Within each chromosome, the regions containing significant associations are represented in more detail (E, F, G, respectively). Lead SNPs identified in the conditional analysis (blue: rs10175462; rs27069; rs9272050; rs6938453; rs55986091; rs9266183) and derived from FinnGen r5 replication (red: rs10175462 or 113992800_TCC_T; rs27069; rs9272050) are represented.

GWAS

Table

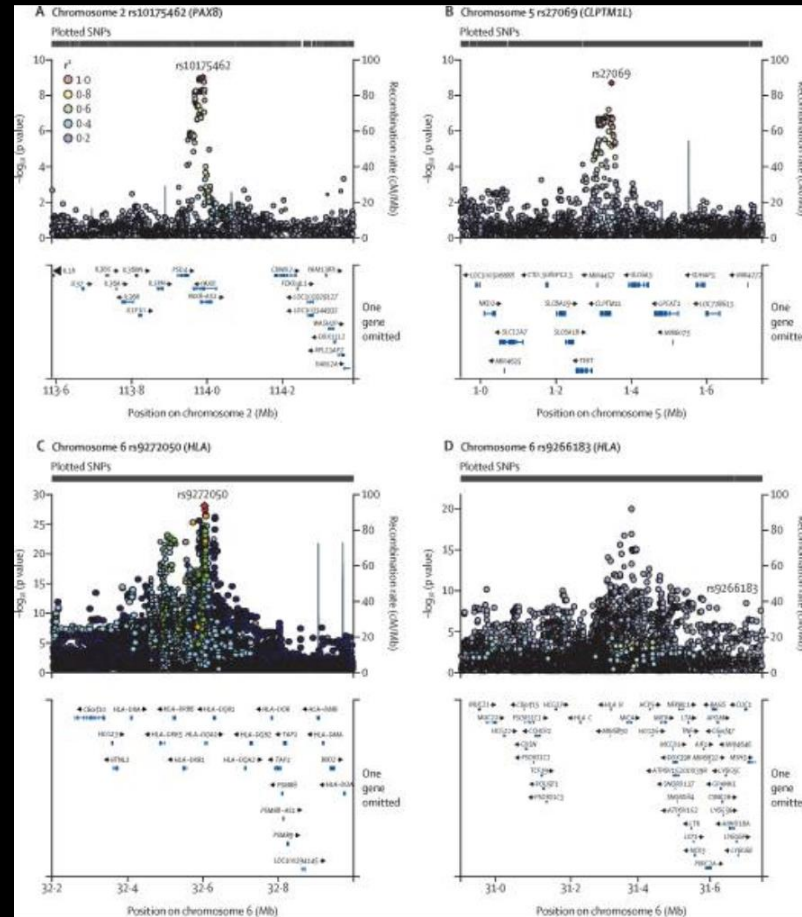
Results from logistic models predicting CIN3 or invasive cervical cancer case-control status from each single-nucleotide polymorphism

	Chromosome	Base pairs	Allele I	MAF	Info	Variant type	Nearest gene	CIN3 or invasive cervical cancer			CIN3			Invasive cervical cancer		
								N	OR (95%CI)	P (Pcond)	N	OR (95%CI)	P (Pcond)	N	OR (95%CI)	P (Pcond)
CIN3 and invasive cervical cancer																
rs10175462	2	113 988 492	A	0.362	1.000	Intron	PAX8	150 314	0.87 (0.84–0.91)	1.07×10^{-9} (1.07×10^{-9})	146 313	0.88 (0.79–0.98)	1.55×10^{-2} (..)	146 309	0.91 (0.82–1.02)	9.26×10^{-2} (..)
rs27069	5	1 347 128	T	0.431	0.986	Intergenic	CLPTMIL	146 530	0.88 (0.84–0.92)	2.51×10^{-9} (2.51×10^{-9})	142 644	0.86 (0.77–0.95)	4.64×10^{-3} (..)	142 641	0.86 (0.77–0.95)	3.61×10^{-3} (..)
rs9266183	6	31 324 647	C	0.037	1.000	Missense	HLA-B	150 314	0.73 (0.64–0.83)	1.53×10^{-6} (6.20×10^{-9})	146 313	0.75 (0.55–1.03)	7.73×10^{-2} (..)	146 309	0.63 (0.45–0.89)	7.88×10^{-3} (..)
rs6938453	6	31 377 793	A	0.228	0.982	Intron	MICA	146 033	0.79 (0.75–0.83)	1.97×10^{-17} (1.35×10^{-15})	142 125	0.79 (0.69–0.91)	7.42×10^{-4} (..)	142 121	0.76 (0.66–0.87)	7.78×10^{-5} (..)
rs9272050	6	32 599 071	G	0.385	0.999	Intron	HLA-DQA1	150 211	1.27 (1.21–1.32)	2.51×10^{-28} (2.51×10^{-28})	146 212	1.42 (1.29–1.58)	9.56×10^{-12} (..)	146 209	1.20 (1.08–1.33)	6.69×10^{-4} (..)
rs55986091	6	32 623 713	A	0.091	0.995	Intergenic	HLA-DQB1	149 757	0.66 (0.60–0.72)	6.42×10^{-22} (1.02×10^{-11})	145 773	0.68 (0.56–0.84)	3.22×10^{-4} (..)	145 771	0.72 (0.59–0.89)	1.84×10^{-3} (..)
CIN3																
rs150406145	2	6 371 190	A	0.012	0.929	Intron	PAX8	149 360	1.33 (1.12–1.56)	8.54×10^{-4} (..)	145 385	2.37 (1.74–3.22)	4.14×10^{-8} (4.14×10^{-8})	145 380	1.05 (0.66–1.65)	8.51×10^{-1} (..)
rs9272245	6	32 602 872	C	0.364	–	Intergenic	HLA-DQA1	149 283	1.26 (1.21–1.31)	1.59×10^{-26} (..)	145 318	1.44 (1.30–1.60)	4.60×10^{-12} (4.60×10^{-12})	145 323	1.21 (1.09–1.34)	4.23×10^{-4} (..)

Models are adjusted for age, smoking status, Townsend index, batch, and the first ten principal components. Results are reported for genome-wide association study significant single-nucleotide polymorphisms following conditional analysis ($p_{\text{cond}} < 5 \times 10^{-8}$) for both CIN3 and invasive cervical cancer, CIN3 only, and invasive cervical cancer only. CIN3=cervical intraepithelial neoplasia grade 3. MAF=minor allele frequency. OR=odds ratio. Pcond=p value after conditional analysis.

GWAS

3 main loci (some replicated in FinnGen)



Functional annotation

Functional analyses of variants and genes

To explore the phenotypic and functional effects of detected genetic loci, we used publicly available databases ([Phenoscaner](#), [UCSC genome browser](#), [dbSNP](#), and [ENCODE](#)) to annotate variants. We searched for previously reported associations with other disease traits in the GWAS Catalog. We subsequently used publicly available datasets to explore potential carcinogenic pathways through the examination of gene expression in cervical cancer tissue, normal cervical tissue, and other tumour tissues (including expression quantitative trait loci [eQTL] ratios in the [PanCanQTL database](#) and the [GTEx Portal](#)). To establish if any gene already had a known role in cancer, we further searched somatic mutations ([COSMIC](#)) in cervical and other cancers. Intronic variants were further investigated for possible indirect effects on gene expression through transcription regulation including histone marks and motifs in [HaploReg v4.1](#) ([appendix p 5](#)).

Functional annotation

Table

Results from logistic models predicting CIN3 or invasive cervical cancer case-control status from each single-nucleotide polymorphism

	Chromosome	Base pairs	Allele I	MAF	Info	Variant type	Nearest gene	CIN3 or invasive cervical cancer			CIN3			Invasive cervical cancer		
								N	OR (95%CI)	P (Pcond)	N	OR (95%CI)	P (Pcond)	N	OR (95%CI)	P (Pcond)
CIN3 and invasive cervical cancer																
rs10175462	2	113 988 492	A	0.362	1.000	Intron	PAX8	150	0.87	1.07×10^{-9}	146	0.88	1.55×10^{-2} (..)	146	0.91	9.26×10^{-2} (..)
								314	(0.84–0.91)	(1.07×10^{-9})	313	(0.79–0.98)		309	(0.82–1.02)	
rs27069	5	1 347 128	T	0.431	0.986	Intergenic	CLPTMIL	146	0.88	2.51×10^{-9}	142	0.86	4.64×10^{-3} (..)	142	0.86	3.61×10^{-3} (..)
								530	(0.84–0.92)	(2.51×10^{-9})	644	(0.77–0.95)		641	(0.77–0.95)	
rs9266183	6	31 324 647	C	0.037	1.000	Missense	HLA-B	150	0.73	1.53×10^{-6}	146	0.75	7.73×10^{-2} (..)	146	0.63	7.88×10^{-3} (..)
								314	(0.64–0.83)	(6.20×10^{-9})	313	(0.55–1.03)		309	(0.45–0.89)	
rs6938453	6	31 377 793	A	0.228	0.982	Intron	MICA	146	0.79	1.97×10^{-17}	142	0.79	7.42×10^{-4} (..)	142	0.76	7.78×10^{-5} (..)
								033	(0.75–0.83)	(1.35×10^{-15})	125	(0.69–0.91)		121	(0.66–0.87)	
rs9272050	6	32 599 071	G	0.385	0.999	Intron	HLA-DQA1	150	1.27	2.51×10^{-28}	146	1.42	9.56×10^{-12} (..)	146	1.20	6.69×10^{-4} (..)
								211	(1.21–1.32)	(2.51×10^{-28})	212	(1.29–1.58)		209	(1.08–1.33)	
rs55986091	6	32 623 713	A	0.091	0.995	Intergenic	HLA-DQB1	149	0.66	6.42×10^{-22}	145	0.68	3.22×10^{-4} (..)	145	0.72	1.84×10^{-3} (..)
								757	(0.60–0.72)	(1.02×10^{-11})	773	(0.56–0.84)		771	(0.59–0.89)	
CIN3																
rs150406145	2	6 371 190	A	0.012	0.929	Intron	PAX8	149	1.33	8.54×10^{-4} (..)	145	2.37	4.14×10^{-8}	145	1.05	8.51×10^{-1} (..)
								360	(1.12–1.56)		385	(1.74–3.22)	(4.14×10^{-8})	380	(0.66–1.65)	
rs9272245	6	32 602 872	C	0.364	–	Intergenic	HLA-DQA1	149	1.26	1.59×10^{-26} (..)	145	1.44	4.60×10^{-12}	145	1.21	4.23×10^{-4} (..)
								283	(1.21–1.31)		318	(1.30–1.60)	(4.60×10^{-12})	323	(1.09–1.34)	

Models are adjusted for age, smoking status, Townsend index, batch, and the first ten principal components. Results are reported for genome-wide association study significant single-nucleotide polymorphisms following conditional analysis ($p_{\text{cond}} < 5 \times 10^{-8}$) for both CIN3 and invasive cervical cancer, CIN3 only, and invasive cervical cancer only. CIN3=cervical intraepithelial neoplasia grade 3. MAF=minor allele frequency. OR=odds ratio. Pcond=p value after conditional analysis.

Functional annotation

Literature searches of genes/variants (some seen in prior GWAS of cervical cancer)

For HLA loci:

“Although we present four independent SNPS that have not been previously reported, linkage disequilibrium patterns from our analysis suggest high linkage with known variants.”

“Of interest is our novel report of the association at a rarer SNP rs9266183 (MAF 0.037; *HLA-B*), which represents a missense variant resulting in a codon shift and amino acid change. Alteration of this allele might result in a shift in peptide presentation properties, distorting the T-cell response to HPV infection.¹⁸ Owing to the lower MAF, this association has probably gone undetected in previous smaller studies. Despite the growing number of *HLA* associations reported for cervical cancer, minimal laboratory validation has been done; given an anticipated coding effect, rs926618 would be one of the most interesting targets for further functional analyses.”

Functional annotation

We identified novel variants in *PAX8* and *CLPTMIL* intronic regions in UK Biobank samples, which were replicated in the FinnGen dataset of cervical cancer phenotypes. Our functional analysis showed how these associations might represent carcinogenic or immune function pathways. The strongest association at the *PAX8* locus was an indel (del:rs35724515), which was in perfect linkage disequilibrium with the lead SNP rs10175462. Enrichment for rs10175462 in cervical cancer tissue was colocalised with an eQTL, and the strongest effects were observed at LOC654433, suggesting long-range epigenetic regulation silencing might be controlling expression.²² Specifically, decreased expression of the whole region was associated with a protective effect in the presence of the alternative T allele. Variant rs10175462 might be tagging the deletion variant rs35724515, resulting in decreased *PAX8* expression leading to a protective effect due to reduced uncontrolled cell growth and anti-apoptosis associated with *PAX8* overexpression.^{23, 24}

Functional annotation

Genetic variation detected at the *CLPTMIL* locus offers additional insight into potential carcinogenic pathways in HPV-driven cancers. *CLPTMIL* codes for a transmembrane protein and has been linked to cell growth promotion in non-cervical tumours,^{[25](#)} and altering cisplatin-mediated apoptosis when overexpressed.^{[26](#)} *CLPTMIL*'s previous association with multiple mucosal and HPV-driven cancers suggests a role in regulation of viral transmission across epithelial barriers. Additional investigations of anti-*CLPTMIL* monoclonal antibodies in chemoresistant lung and pancreatic cancers suggest that *CLPTMIL* is a potential novel target for cancer therapeutics in various cancers.^{[27](#)}

Mendelian randomization

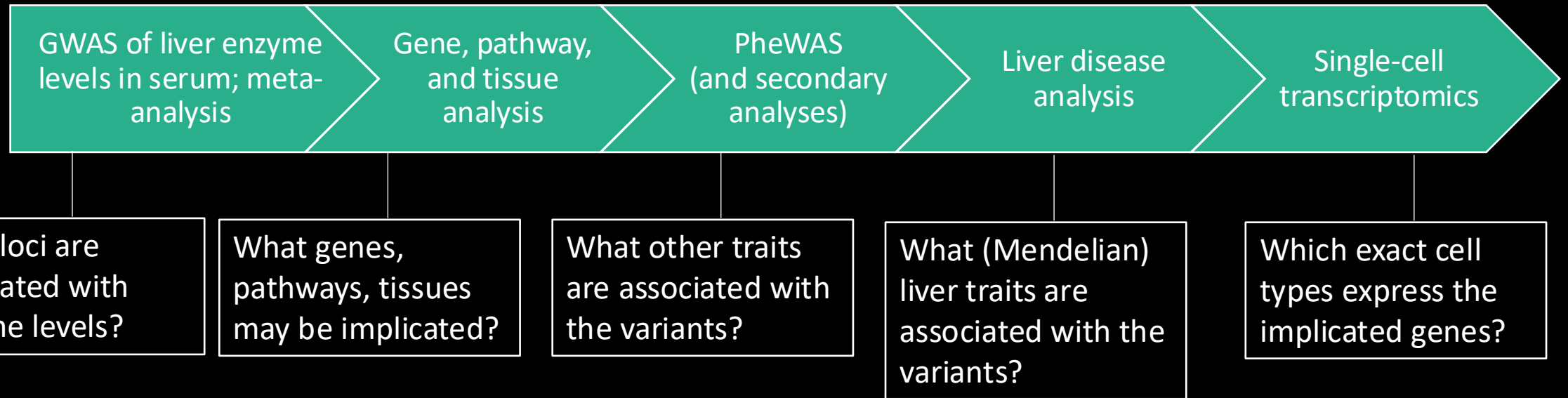
“[Two-sample] Mendelian randomisation further supported the complementary role of smoking (OR 2.46, 95% CI 1.64–3.69), older age at first pregnancy (0.80, 0.68–0.95), and number of sexual partners (1.95, 1.44–2.63) in the risk of developing cervical cancer.”

GWAS #4

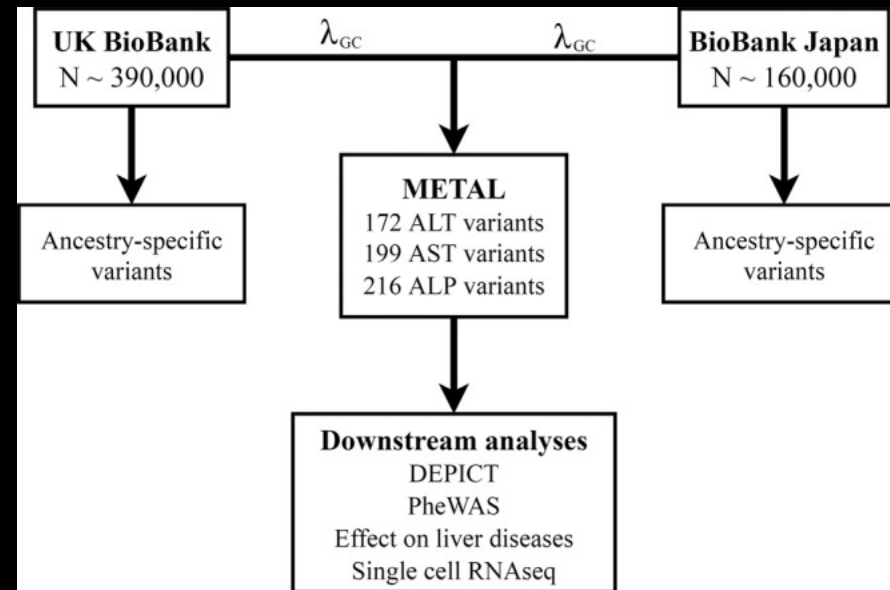
Follow-up approach:

Focus on phenotypes/expression
in relevant tissues

Chen, V. L., Du, X., Chen, Y., Kuppa, A., Handelman, S. K., Vohnoutka, R. B., Peyser, P. A., Palmer, N. D., Bielak, L. F., Halligan, B., & Speliotes, E. K. (2021). Genome-wide association study of serum liver enzymes implicates diverse metabolic and liver pathology. *Nature communications*, 12(1), 816.
<https://doi.org/10.1038/s41467-020-20870-1>



GWAS



“Liver enzymes such as alanine transaminase (ALT), aspartate transaminase (AST), and alkaline phosphatase (ALP) are the most commonly-used laboratory indicators of liver disease and are increased in most common liver diseases”

GWAS

We defined/identified
172 independent ALT,
199 AST, and
216 ALP loci
after eliminating any SNPs within 1 Mb or
LD ($R^2 > 0.01$) of another genome-wide
significant locus for the same trait

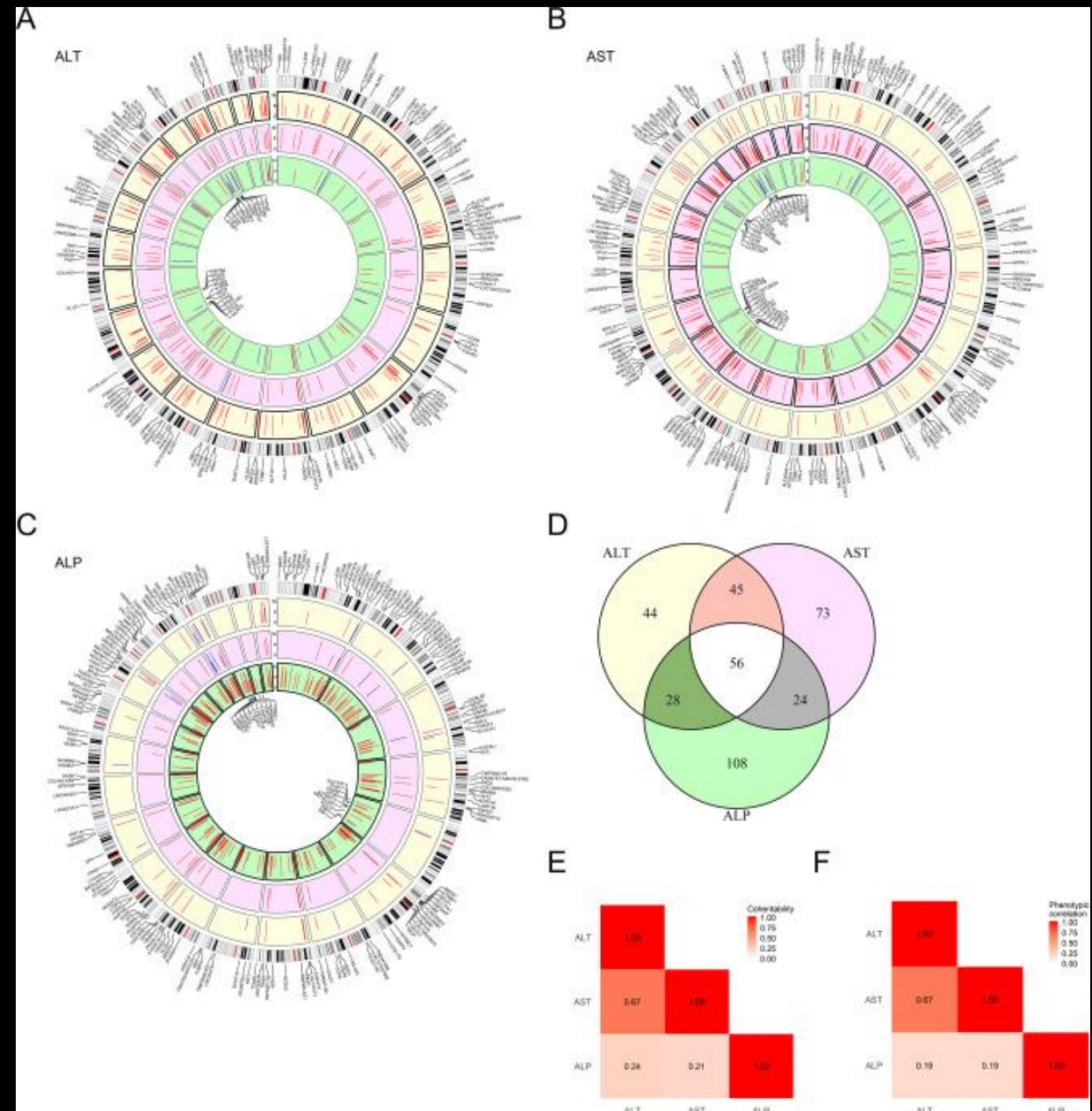
Genetic loci associated with liver enzymes

A–C Circos plots depicting variants (unadjusted $p < 5 \times 10^{-8}$) with **A** alanine transaminase (ALT), **B** aspartate transaminase (AST), and **C** alkaline phosphatase (ALP). Outermost circles represent associations with ALT, middle AST, innermost ALP.

D overlap of loci across traits.

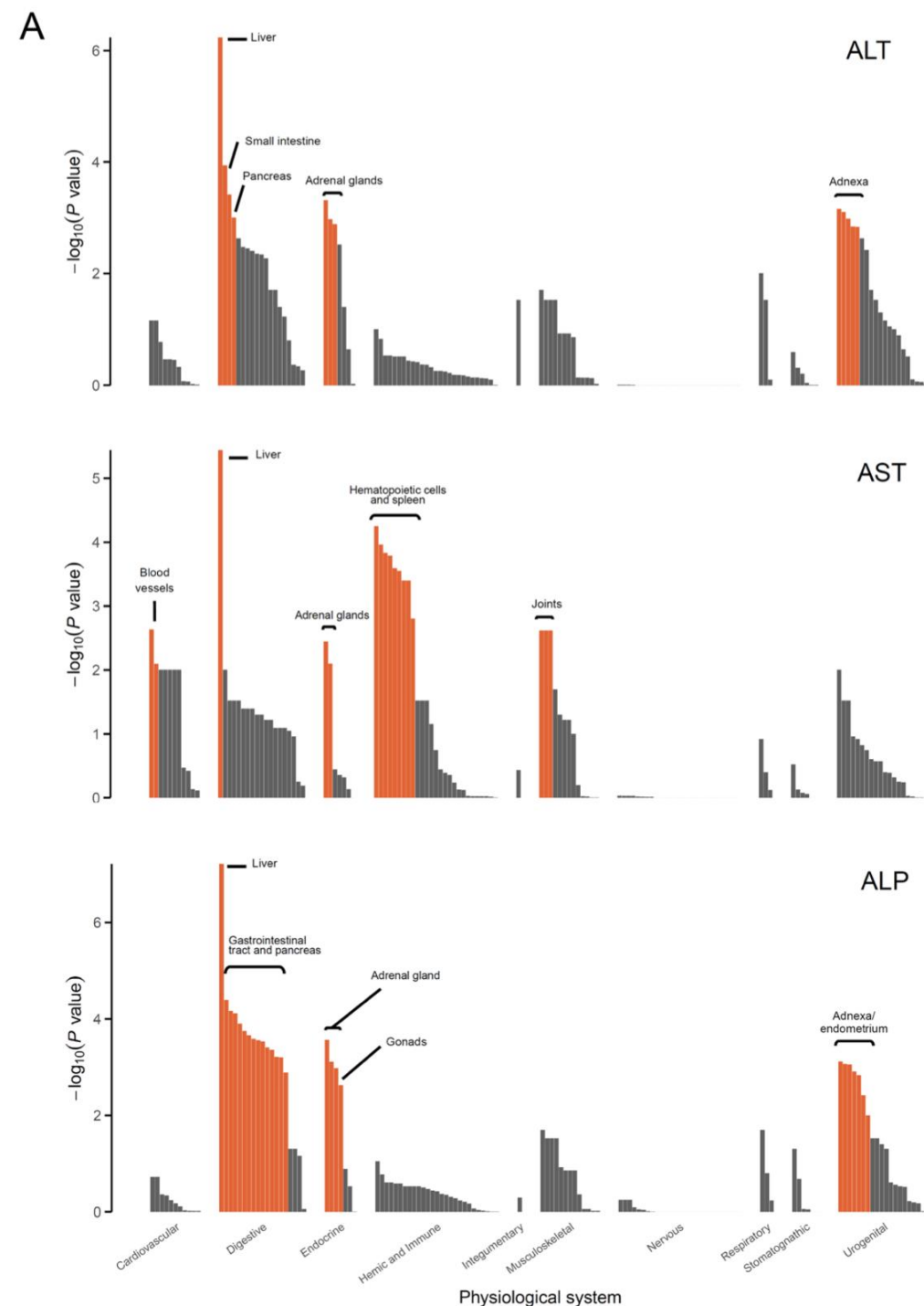
E Co-heritability of ALT, AST, and ALP.

F Phenotypic correlation of ALT, AST, and ALP.



Gene, pathway, tissue analysis (DEPICT)

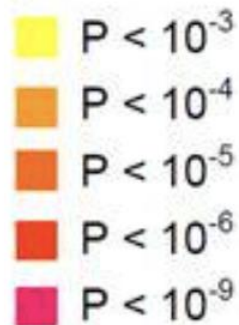
We used DEPICT, a program that uses GWAS-prioritized genes and gene co-expression patterns across cells and tissues, to identify tissues, pathways, and genes (using an FDR < 0.05) that are enriched for associations with a trait



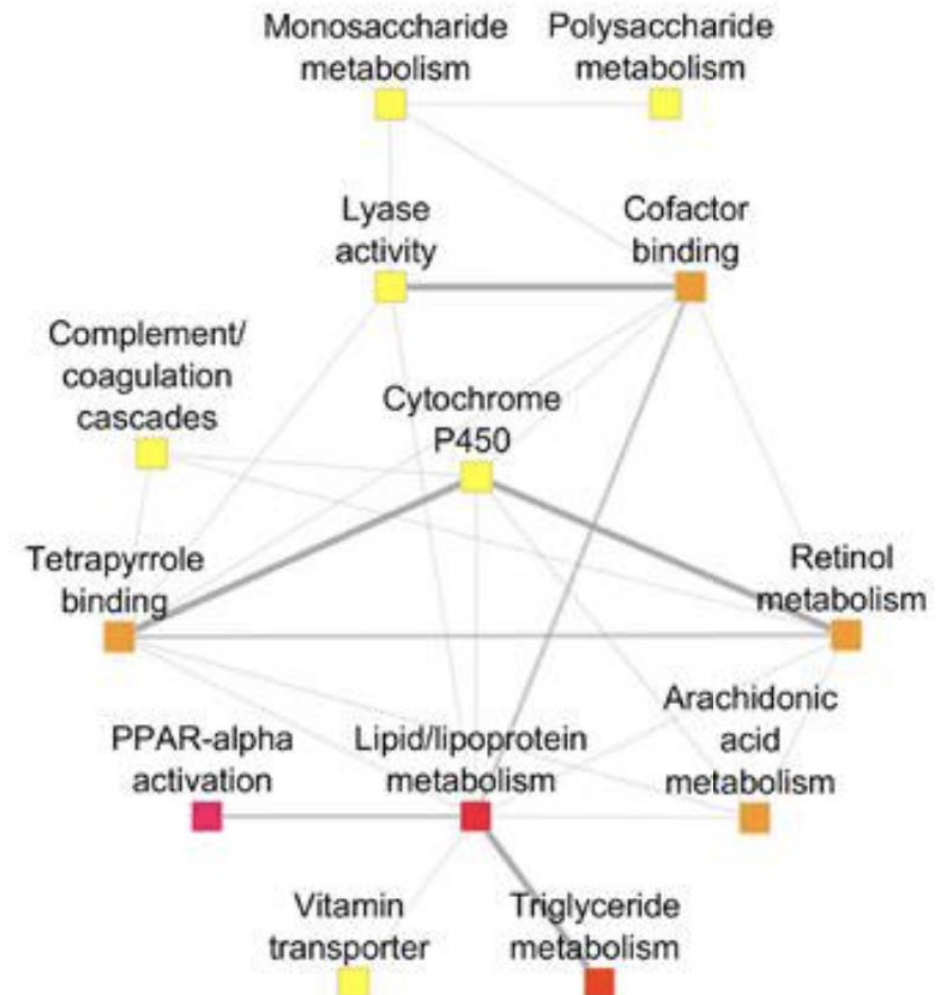
Gene, pathway, tissue analysis (DEPICT)

Network plots depicting gene sets enriched in all three sets of ALT-, AST-, and ALP-associated variants (“Common”)

FDR-corrected
p values

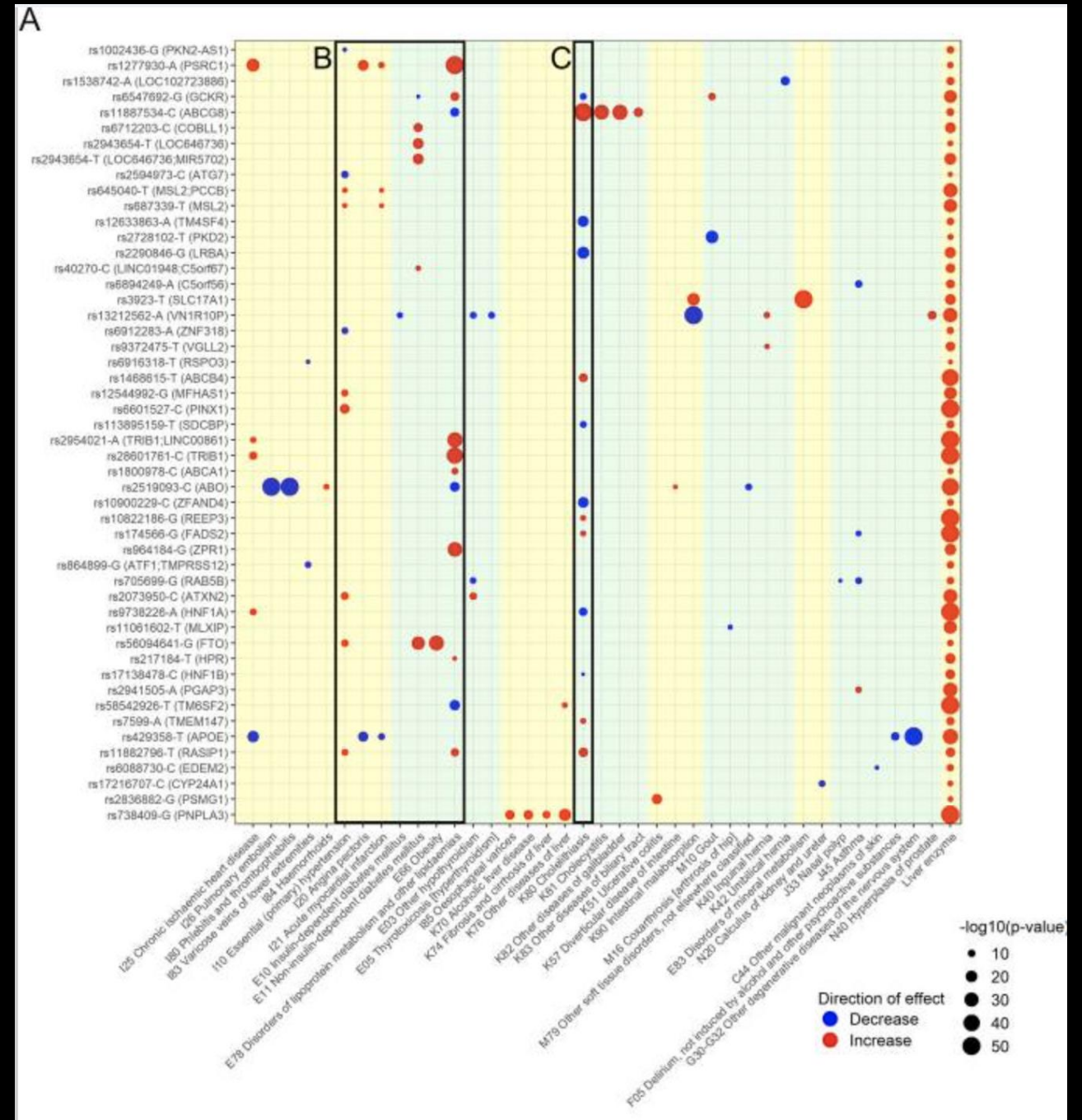


Common



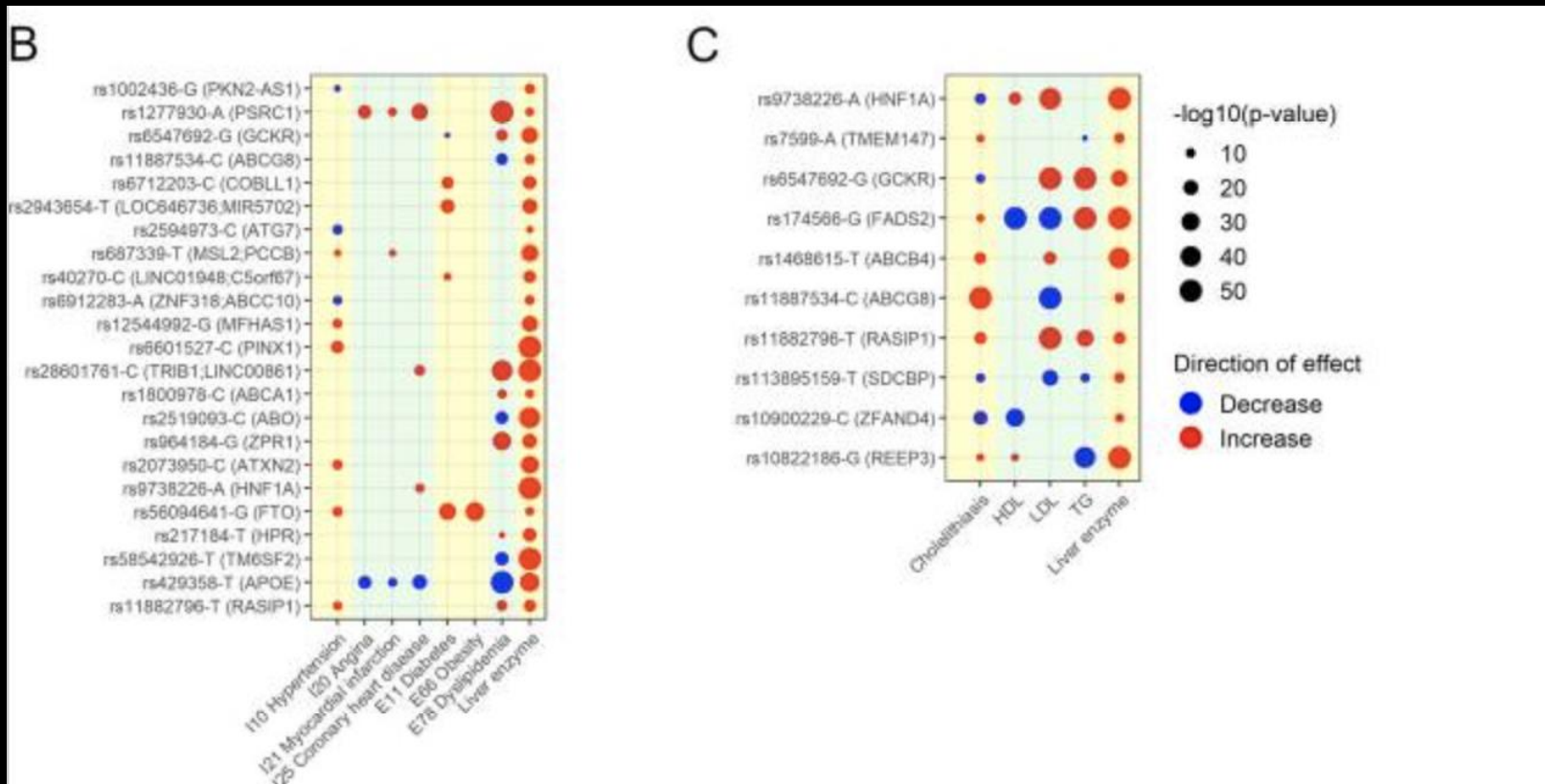
PheWAS

International Classification of Diseases (ICD)-
based diseases using previously-published
PheWAS data



PheWAS

Notable findings were that a number of alleles were associated with metabolic diseases and cholelithiasis. 23 alleles associated with metabolic traits including hypertension, dyslipidemia, diabetes, and coronary artery disease (Fig. 4B). 13 alleles associated with variations in risk of cholelithiasis (Fig. 4C). Several alleles associated with other traits of interest: rs2836882 near *PSMG1* associated with ulcerative colitis, multiple SNPs including rs3923 near *SLC17A1* associated with disorders in mineral metabolism, and rs17216707 near *CYP24A1* associated with nephrolithiasis.



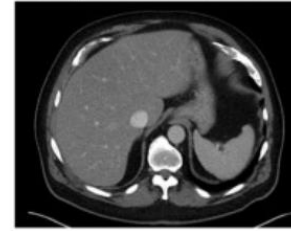
Further analyses sought to disentangle causality in these associations:

“There was no significant evidence for causality of ALT, AST, or ALP with any metabolic traits. These analyses suggest that liver enzyme elevations themselves do not cause metabolic diseases, but instead are the result of metabolic disease such as obesity and dyslipidemia.”

Overlap with previous liver disease GWAS

Associations of liver enzyme-increasing genetic variants on liver diseases. The nearest gene to each variant is depicted. Red text indicates that the liver enzyme-increasing allele increases risk of that disease, while blue text indicates that it lowers risk of that disease (Bonferroni-corrected). Black text indicates a trend toward a significant effect ($p < 0.05$). When two or three variants < 1 MB apart are associated with different liver enzymes are also associated with a liver trait, this was considered a single genetic locus and the variant with the lowest p -value for its respective liver enzyme was used. These variants are highlighted with * indicating that the locus had opposite directions of association on the liver enzymes, or with + indicating that the locus had the same direction of association on the liver enzymes. Hepatic steatosis is defined as negative computed tomography-measured liver attenuation, so that an allele that is associated with increased liver attenuation would be associated with decreased hepatic steatosis, and vice versa.

Hepatic steatosis



PNPLA3+, TM6SF2*

AGPAT5+, AKR1C4, COL4A2, CCDC126, ERLIN1+, FGF9, GCKR, GPAM+, HNF1A, LCORL, LINC00243, LOC102723704, LRBA, MARC1+, MIR4713HG+, MLXIP, PEPD, RAB5B, SEMA4B, TMEM263, TRIB1

LOC157273+

APOB, ARID1A, ATF1, FAM47E-STBD1, HARS2, LINC00184, MGAT5, RIC1, ST5, TARS2, TNKS2

Alcoholic cirrhosis



TM6SF2*, PNPLA3+, MBOAT7

EBF1, HNF1A, HPR, LOC101927668, MARC1+, MIR5705+, NAP1L4, NIN, RNF213, SOX7, ST3GAL4, TM4SF4, TNFRSF11A, TOR1B, TRIM59+

ABO*, CDKN1A, CFH, ERAP2, FBXO11, GLB1, MLH1, NINJ1+, REEP3, WDR91

Primary sclerosing cholangitis



Click on image to zoom

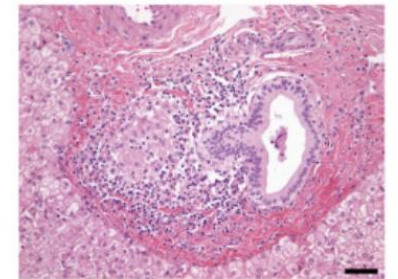
PSMG1, SLC17A1

ABCG8, ATXN2+, BMP8A, CASP10, CTSS, EDEM2, H3F3B, HNF1B+, ITGA1+, LARP1B, LINC01336, LINC02341, LOC157273+, LOC102724084+, MAFF*, NCOA2, NT5DC1+, MCM6, RASIP1, STRN, TET2, TMBIM1, TRIM5+, VIM-AS1

LINC00243, OR2H2, PRKD2, VN1R10P

ABO, DEF6+, FLJ45513, FUBP3, GLDC, LINC01923, MAFTTR, MIR4713HG, MLXIP, PTCH1, TMEM150B, STARD10, UBE2E3, USP43, ZFAND4, ZPR1

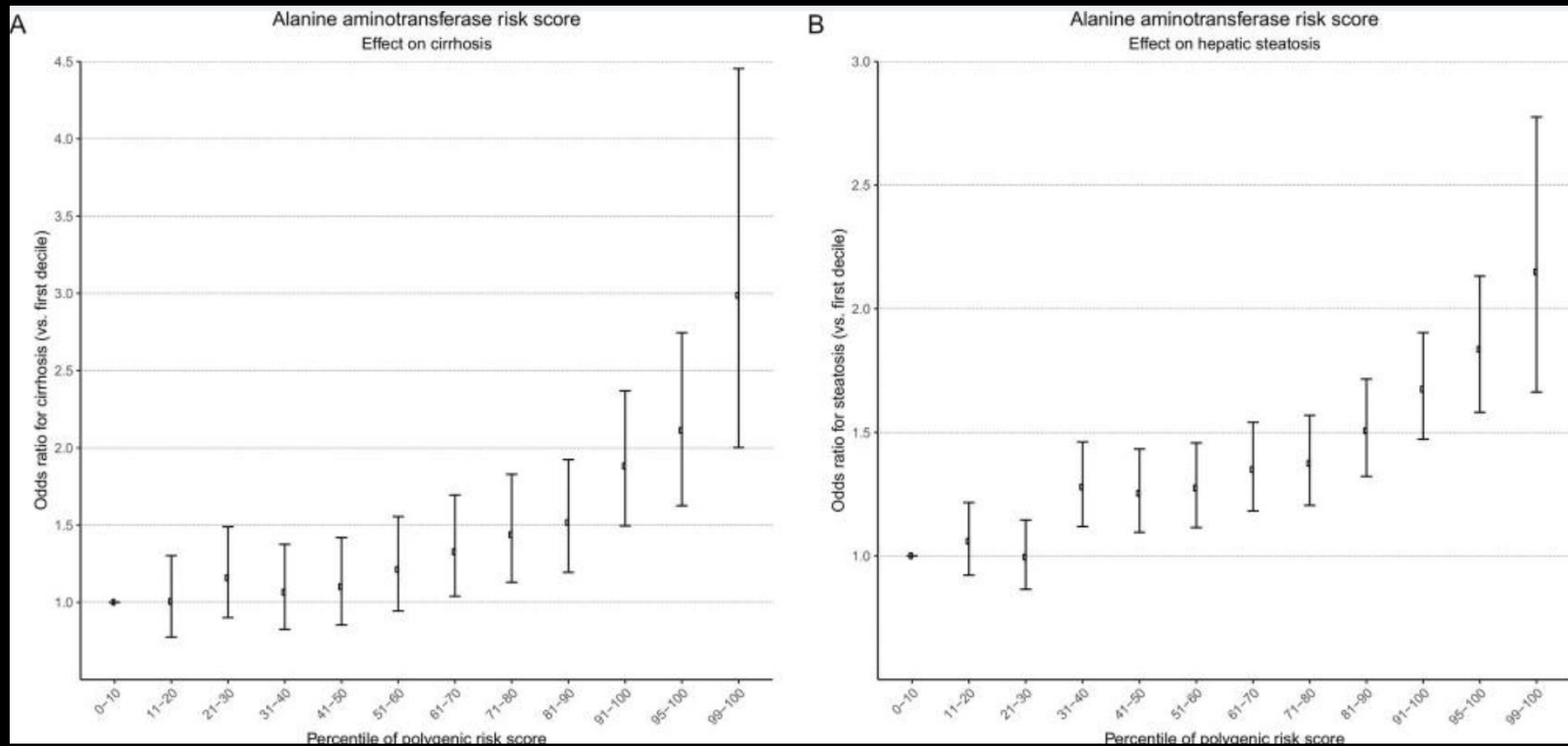
Primary biliary cholangitis



TRIM59

MANBA

PRS



Overlap with Mendelian disease

Next, we investigated associations between non-synonymous mutations in LD with the liver enzyme-associated variants and Mendelian diseases using the Online Mendelian Inheritance in Man catalog

“...Other potentially deleterious missense mutations in LD with ALT-increasing alleles were found in *ANO5*, which is associated with Miyoshi muscular dystrophy; the breast/ovarian cancer susceptibility gene *BRCA1*; and *LRBA*, associated with common variable immunodeficiency. These LD exonic alleles were associated with abnormal liver enzymes even after adjustment for previously-reported Mendelian disease variants and constitute new physiology-altering if not disease-causing variants.”

Table 1

Genetic variants associated with both liver enzyme levels and Mendelian disease genes.

Disease	Mendelian disease gene	Liver enzyme lead SNP	Exonic SNP	Exonic SNP deleteriousness			
				CP	Sift	HD	HV
<i>Sitosterolemia</i>	<i>ABCG8</i>	<i>rs11887534</i>	<i>rs11887534</i>	22.7	D	P	P
<i>Sitosterolemia</i>	<i>ABCG5</i>	<i>rs11887534</i>	<i>rs6756629</i>	23.0	D	D	P
Gnathodiaphyseal dysplasia; Miyoshi muscular dystrophy 3	ANO5	rs7481951	rs7481951	12.1	T	B	B
Carbamoyl phosphate synthetase I deficiency	CPS1	rs1047891	rs1047891	22.1	T	B	B
Lipoid proteinosis of Urbach and Wiethe	ECM1	rs1815544	rs3737240;	0.1;	T;	D;	P;
			rs13294	9.4	T	D	B
Spastic paraplegia	ERLIN1	rs2862954	rs2862954	13.3	T	B	B
Deafness	S1PR2	rs10409243	rs2116942	NA	NA	NA	NA
Ceroid lipofuscinosis, neuronal, 10	CTSD	rs11555039	rs17571	1.5	T	B	B
Breast-ovarian cancer, familial; Fanconi anemia, complementation group S	BRCA1	rs8176279	rs1799966	0.5	T	P	B
Retinitis pigmentosa	MERTK	rs6541998	rs7604639; rs2230515	0.0	T	B	B
Malignant hyperthermia;Hypokalemic periodic paralysis; Thyrotoxic periodic paralysis	CACNA1S	rs3850625	rs3850625	34.0	D	P	B
Brain small vessel disease	COL4A2	rs9521787	rs7990383	0.0	T	B	B
Ellis-van Creveld syndrome; Weyers acrofacial dysostosis	EVC	rs2279252	rs2279252	23.0	T	P	B
Immunodeficiency, common variable	IRF2BP2	rs599633	rs7545855	0.0	T	B	B
Immunodeficiency, common variable, 8, with autoimmunity	LRBA	rs2290846	rs2290846;	25.9;	T	B	B
			rs3749574	19.9			
Mannosidosis, beta B, lysosomal	MANBA	rs179195	rs2866413	6.5	T	B	B
Encephalitis/encephalopathy, mild, with reversible myelin vacuolization; Cardiac-urogenital syndrome	MYRF	rs174566	rs174535	NA	NA	NA	NA
Lethal congenital contracture syndrome; Nevus comedonicus	NEK9	rs4903273	rs10146482	18.7	T	B	B
Acrodermatitis enteropathica, zinc-deficiency type	SLC39A4	rs13273326	rs2280838;	8.0;	T;	B;	B;
			rs2280839	5.7	D	B	B
Paget disease of bone, juvenile-onset	TNFRSF11B	rs11573824	rs2073618	16.0	T	B	B
Trehalase deficiency	TREH	rs12225548	rs2276065	0.0	T	B	B
Combined oxidative phosphorylation deficiency	VARs2	rs9501030	rs9394021	27.5	D	D	D
Cerebellar ataxia, mental retardation, and dysequilibrium syndrome; Hydrocephalus, congenital, with brain anomalies	WDR81	rs11078597	rs8077638	NA	NA	NA	NA

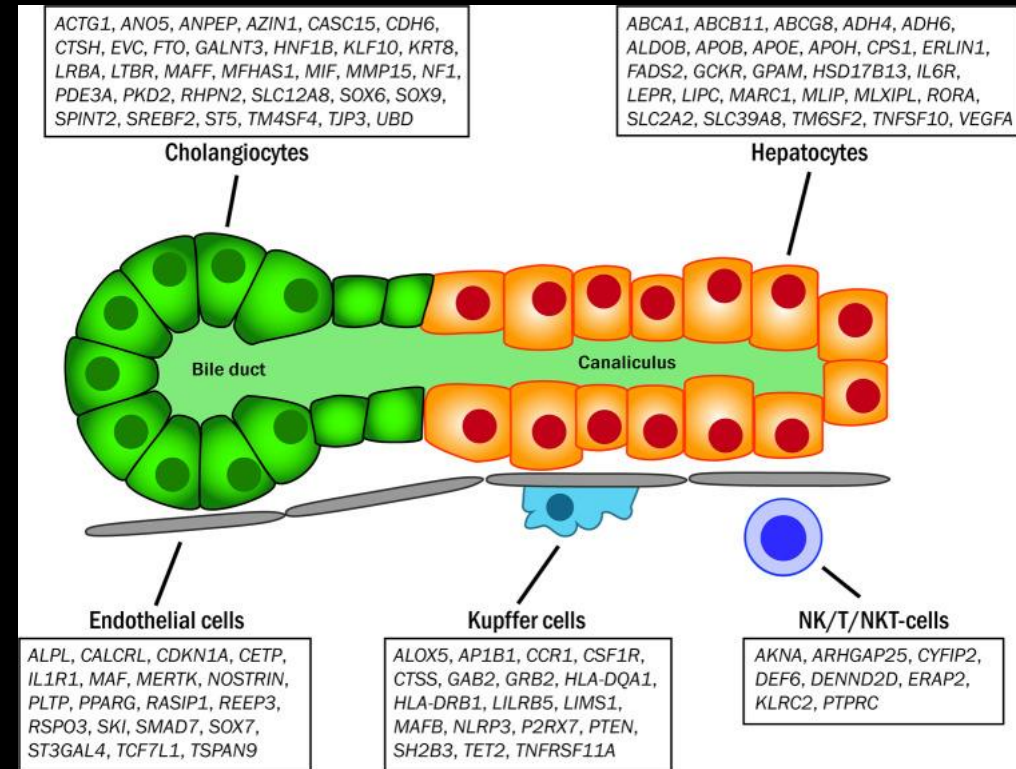
SNP single nucleotide polymorphism, *CP* Combined Annotation-Dependent Depletion-Phred, *HD* HumDiv (PolyPhen), *HV* HumVar (PolyPhen), *NA* not available.

Bold denotes that the liver enzyme-affecting variant influences liver enzymes independently of previously-reported Mendelian disease-causing variants. Italics denotes that the liver enzyme-affecting variant is the same as a previously-reported Mendelian disease-causing variant.

Single-cell RNA-seq

Finally, we used recently-reported human single cell RNA sequencing data from human liver to investigate cell-specific expression of the genes nearest to the liver enzyme-altering variants

“Combining single cell sequencing and GWAS may suggest mechanisms by which a specific variant affects liver disease. For example, *TM6SF2* was expressed exclusively in hepatocytes, consistent with its reported effect on very low-density lipoprotein export in hepatocytes³⁶. *PNPLA3*, in contrast, was expressed at high levels in both hepatocytes and cholangiocytes compared to other cell types.”



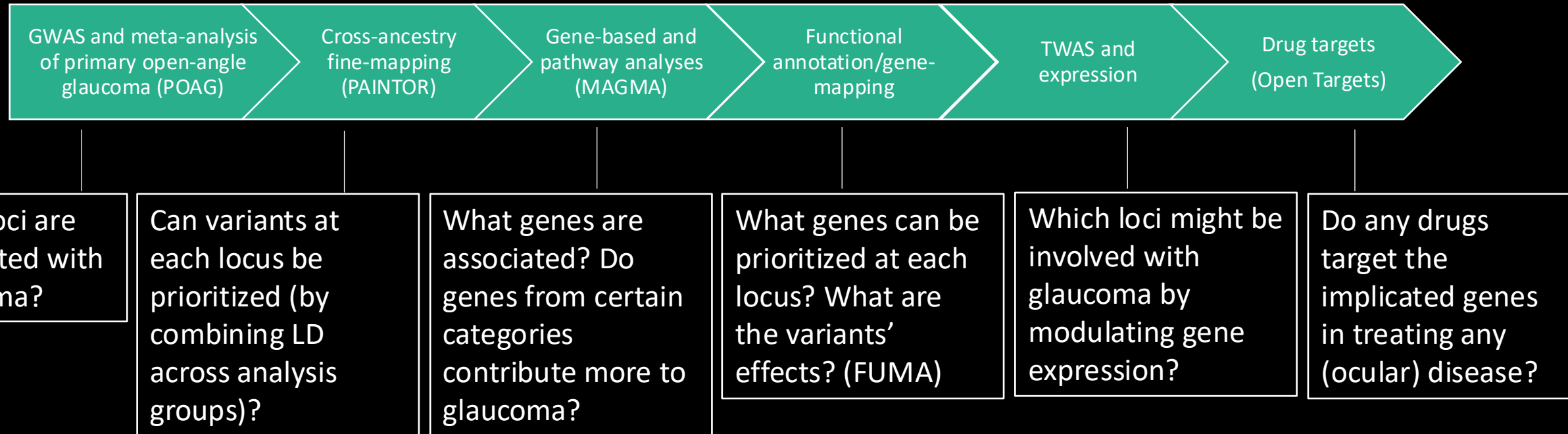
Cell type-specific expression of genes nearest to selected liver enzyme-associated genetic variants
Gene names appear in the boxes corresponding to the cell type in which they are specifically expressed.

Gharahkhani, P., Jorgenson, E., Hysi, P., Khawaja, A. P., Pendergrass, S., Han, X., Ong, J. S., Hewitt, A. W., Segrè, A. V., Rouhana, J. M., Hamel, A. R., Igo, R. P., Jr, Choquet, H., Qassim, A., Josyula, N. S., Cooke Bailey, J. N., Bonnemaier, P., Iglesias, A., Siggs, O. M., Young, T. L., ... Wiggs, J. L. (2021). Genome-wide meta-analysis identifies 127 open-angle glaucoma loci with consistent effect across ancestries. *Nature communications*, 12(1), 1258. <https://doi.org/10.1038/s41467-020-20851-4>

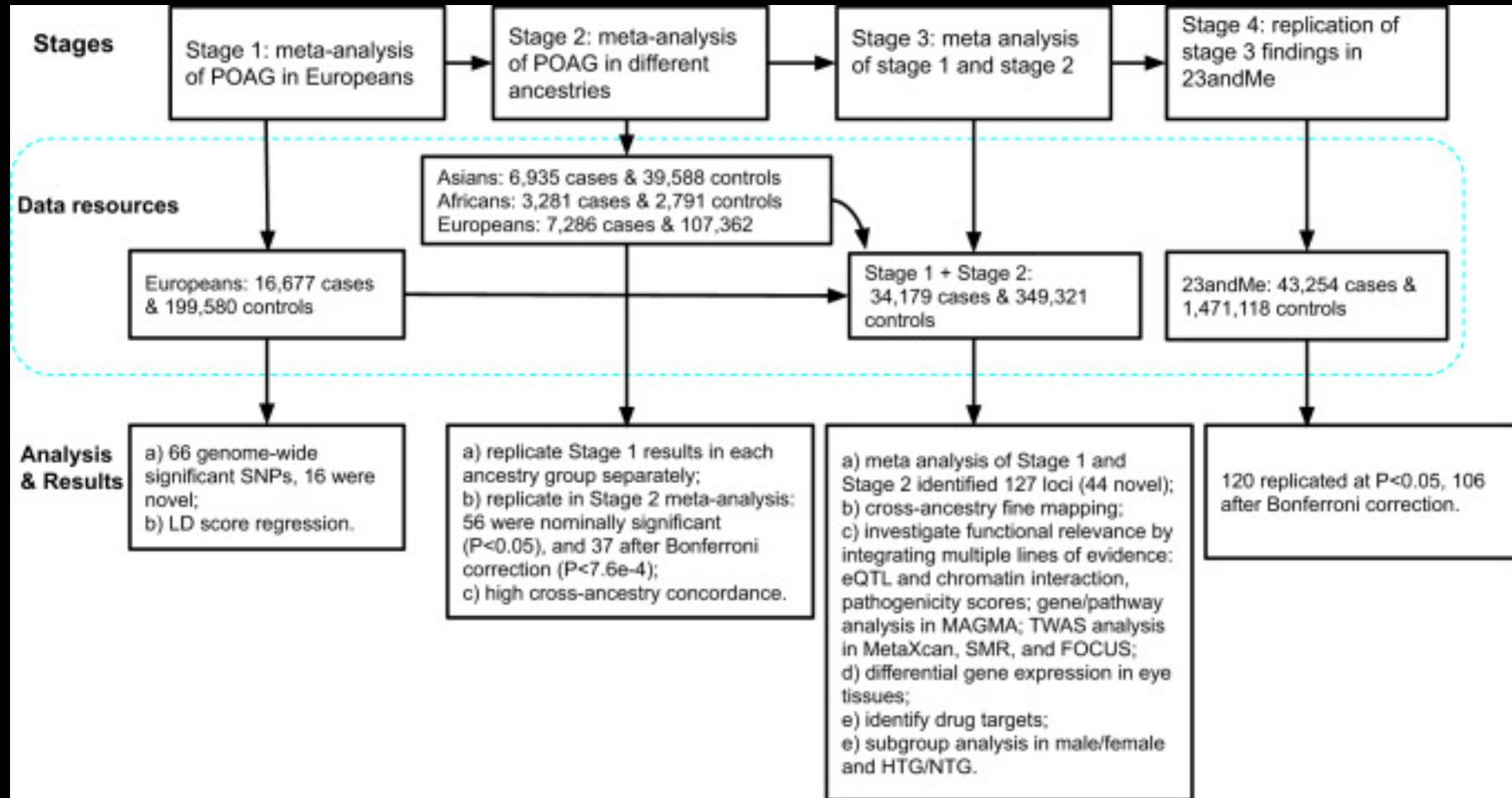
GWAS #5

Follow-up approach:

“Do a lot of stuff and informally integrate it all”

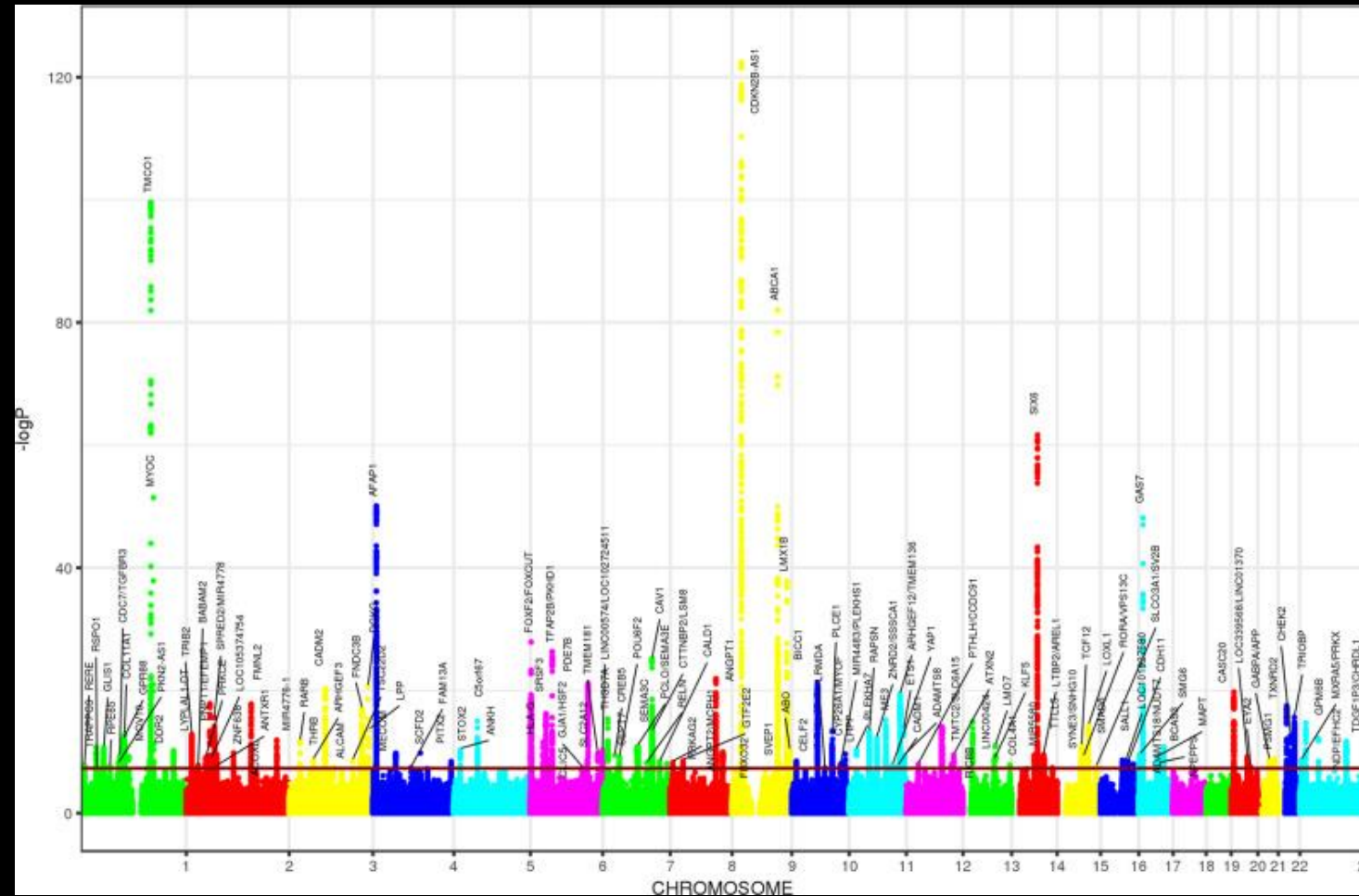


GWAS #5



GWAS

Manhattan plots for the cross-ancestry meta-analysis.

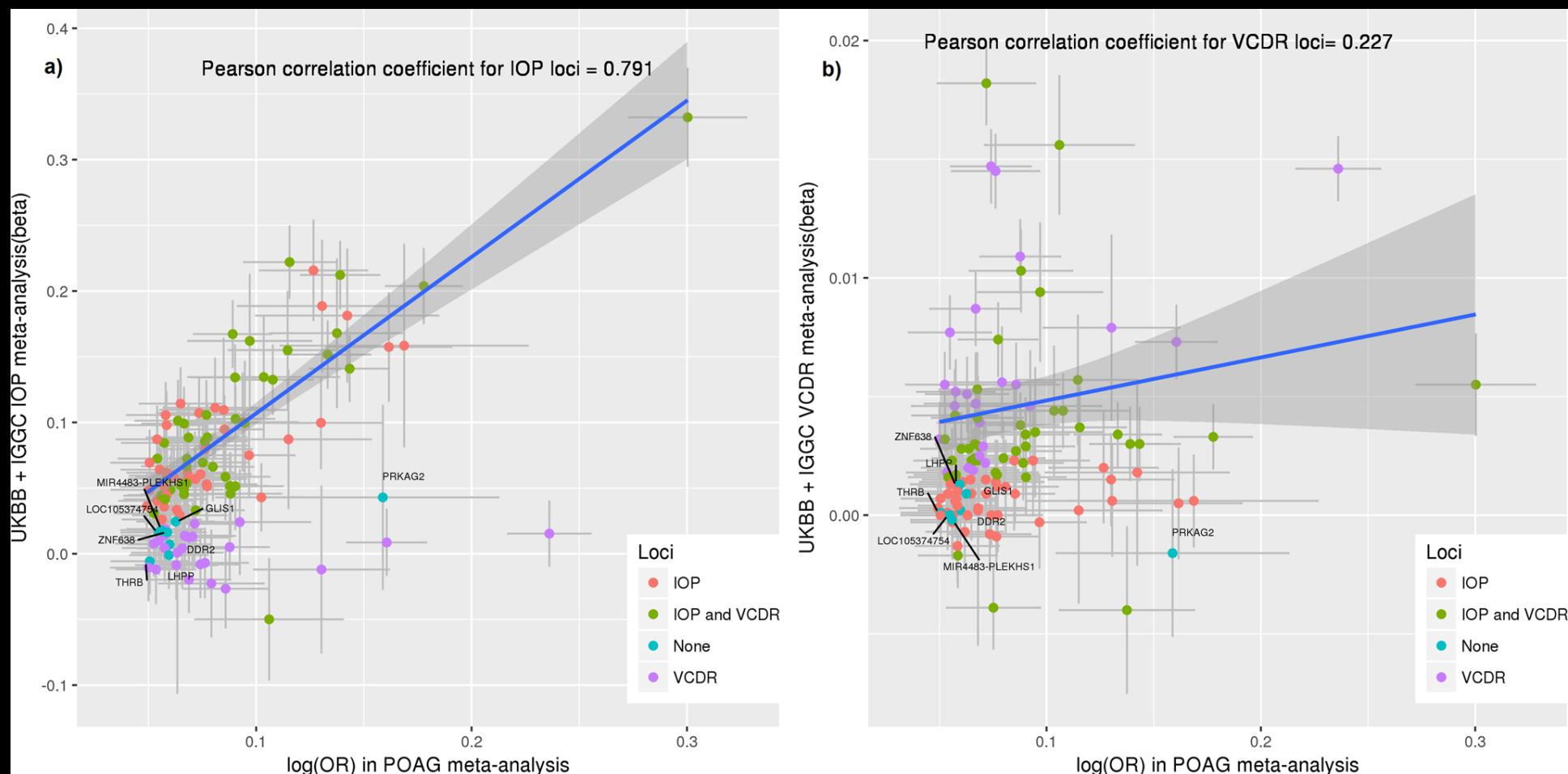


“Pleiotropy”

“All POAG risk variants identified to date have also been associated with either IOP or with VCDR or both. We investigated the association of the POAG loci identified in this study with IOP and VCDR, using previous GWAS data .”

→ “Most of the risk loci associated with POAG involve known glaucoma-related endophenotypes”

“Several highly heritable endophenotypes are related to POAG risk including IOP, structural variation of the optic nerve characterized as vertical cup-to-disc ratio (VCDR) and variation in thickness of the retina cell layers including the retinal nerve fiber layer (RNFL) and the ganglion cell inner plexiform layer (GCIPL)”



Colocalization

“Given that three POAG risk loci that we identified (loci containing *MAPT*, *CADM2*, and *APP*) have also been implicated in Alzheimer’s disease (AD) and dementia^{26,27,28}, we asked whether the same causal variants may underlie these loci and whether there is evidence for genetic sharing in any of the other genome-wide significant POAG loci. Applying the Bayesian-based colocalization method, eCAVIAR²⁹, to the cross-ancestry and European POAG GWAS meta-analyses and the publicly available AD GWAS data³⁰ for 21,982 cases and 41,944 controls³⁰ of European descent, we found no evidence for sharing of causal variants in the 123 (cross-ancestry) or 66 (European) autosomal POAG loci (Colocalization Posterior Probability (CLPP) < 0.01; Supplementary Data 7).”

Fine-mapping

“Overall, the best causal SNPs in Europeans changed for 52 of 127 loci after including data from the other ancestries ”

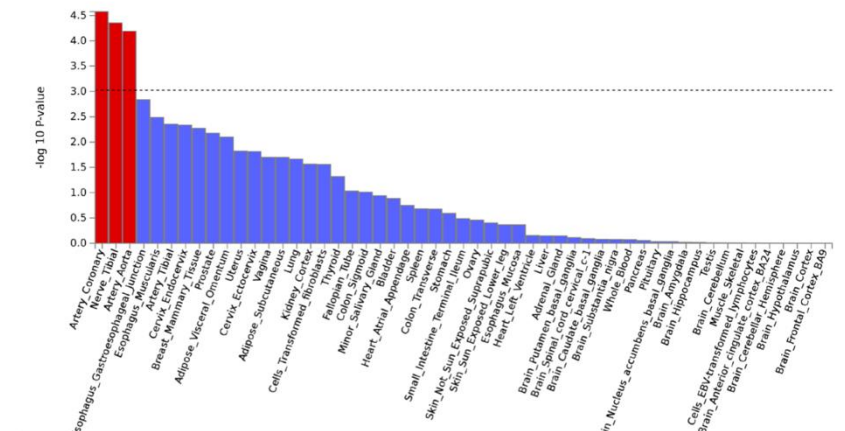
Gene-based and pathway analyses (MAGMA)

Expression of the risk genes identified in MAGMA gene-based analysis were significantly enriched in artery and nerve tissues, reflecting the widely recognized neuronal and vascular character of glaucoma

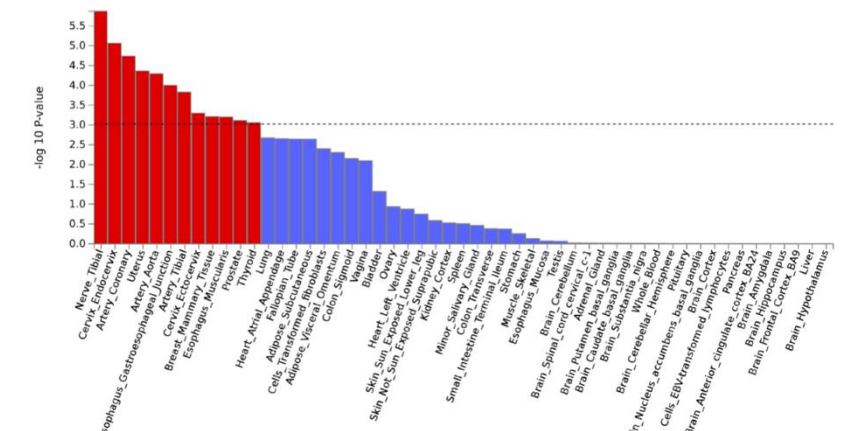
Pathway analysis identified 21 significant gene-sets surviving the Bonferroni-corrected threshold ($P < 0.05/10678$). These included previously identified pathways such as collagen formation and vascular development, and highlighted additional pathways involved in lipid binding and transportation such as apolipoprotein binding and negative regulation of lipid storage

Supplementary Figure 5. Enrichment of the expression of the significant genes identified by MAGMA in GTEX tissues. The horizontal line indicates the significance threshold ($P < 1e-03$), Bonferroni-corrected for 53 GTEX tissues tested.

a) Results from the European meta-analysis.



b) Results from the cross-ancestry meta-analysis.



Functional annotation/gene-mapping

Table 1

Summary of the previously unreported POAG risk loci annotation and functional studies.

Lead SNP	Annotation	Nearest gene	Causal ^a	Chromatin interaction ^b	eQTL any tissue ^c	eQTL retina ^d	Protein altering ^e	CADD > 12.37 ^f	Gene-based tests ^g	Genes within significant pathways ^h
rs172531	Intron	RERE	NO	RPL7P11	RERE	RERE	NO	NO	RERE	NO
rs941125	Intron	GLIS1	NO	SLC25A3P1	GLIS1	NO	NO	NO	GLIS1	NO
rs4076000	Intergenic	ELOCP18/RPE65	NO	CTBP2P8	WLS	NO	NO	NO	NO	NO
rs12566440	Intron	GPR88	NO	VCAM1	VCAM1	VCAM1	NO	NO	GPR88	NO
rs4542196	Intron	DDR2	NO	DDR2	DDR2	NO	NO	NO	NO	DDR2
rs12623251	Intergenic	TRIB2	YES	AC013471.1	TRIB2	NO	NO	NO	TRIB2	NO
rs6713914	Intron	LOC105374754	NO	AC009970.1	NO	NO	NO	YES	NO	NO
rs12613800	Intron	ZNF638	NO	RNU6-105P	ZNF638	ZNF638	ZNF638	NO	ZNF638	NO
rs9852634	Intron	THRB	YES	RPL31P20	NO	NO	NO	NO	NO	NO
rs1500708	Intron	ARHGEF3	YES	ARHGEF3-AS1	NO	NO	NO	NO	ARHGEF3-AS1	NO
rs6437582	Intergenic	ALCAM	YES	AC074043.1	ALCAM	ALCAM	NO	NO	NO	NO
rs10517281	Intron	SCFD2	YES	Multiple ⁱ	FIP1L1	NO	NO	YES	SCFD2	NO
rs57400569	Intron	FAM13A	YES	Multiple	FAM13A	NO	NO	YES	NO	NO
rs17527016	Intergenic	PITX2	NO	AC083795.1	NO	NO	NO	YES	NO	PITX2
rs6552711	Intron	STOX2	NO	ENPP6	NO	NO	NO	NO	NO	NO
rs407238	Intergenic	HLA-G	NO	Multiple	NO	NO	NO	NO	Multiple	NO
rs3777588	Intron	CLIC5	NO	MIR4642	CLIC5	CLIC5	NO	YES	CLIC5	NO
rs7760346	Intergenic	GJA1/HSF2	NO	Multiple	HSF2	Multiple	NO	NO	NO	GJA1
rs2811688	Intron	SLC2A12	NO	TBPL1	SLC2A12	SLC2A12	NO	NO	SLC2A12	NO
rs2191828	Intron	CREB5	NO	AC005105.1	NO	NO	NO	NO	NO	NO
rs6957752	Intergenic	SEPT7	NO	Multiple	RP11-379H18.1	RP11-379H18.1	NO	NO	NO	NO
rs7805468	Intergenic	PCLO/SEMA3E	NO	AC079799.1	NO	NO	NO	NO	SEMA3E	SEMA3E
rs2515437	Intron	ANGPT2/MCPH1	NO	Multiple	NO	NO	ANGPT2	NO	NO	ANGPT2

^aThis column indicates whether the GWAS lead SNP was identified to be the most probable causal SNP across ancestries in the multi-ethnic fine-mapping analyses performed using the software PAINTOR.

^bWhere the GWAS lead SNP or SNPs in LD $r^2 > 0.8$ with the lead SNP overlap with one end of a chromatin interaction, the most significant gene involved in this interaction has been shown.

^cWhere the GWAS lead SNP or SNPs in LD $r^2 > 0.8$ with the lead SNP is an eQTL in any GTEx tissue or the other eQTL databases used in this study (see the Methods), the most significant target gene has been shown.

^dWhere the GWAS lead SNP or SNPs in LD $r^2 > 0.8$ with the lead SNP is an eQTL in retina in EyeGEx study, the most significant target gene has been shown.

^eWhere the GWAS lead SNP is a protein-altering variant or in LD $r^2 > 0.8$ with a protein-altering variant, the corresponding gene has been shown.

^fThis column indicates whether the GWAS lead SNP has a CADD score > 12.37 .

^gThis column shows the loci in which one or multiple genes were significant in any of the gene-based tests (MAGMA, MetaXcan, SMR, and FOCUS) used in this study. “Multiple” indicates that several genes are involved. These genes have been named in Supplementary Data ¹⁴.

^hThis column shows the loci for which the reported nearest genes or the significant genes identified in the gene-based tests above are members of at least one significant POAG pathway identified in this study.

ⁱ“Multiple” indicates several genes.

Functional annotation/gene-mapping

Bioinformatic functional analyses

The bioinformatic functional analysis to investigate the functional relevance of the identified risk loci for POAG were performed through the FUMA platform v1.3.5⁸³ using the following dataset/toolsets: GTEX eQTL v6⁸⁴; Blood eQTL browser⁸⁵; BIOS QTL Browser⁸⁶; BRAINEAC⁸⁷; RegulomeDB v1.1³⁶; CADD v1.4³⁵; ANNOVAR (the version released on 2016Feb01)⁸⁸; and Hi-C data from 21 tissue/cell types (GEO accession: [GSE87112](#))⁸⁹, PsychENCODE⁹⁰, Giusti-Rodriguez et al. (2019)⁹¹, and FANTOM5 Human Enhancer Tracks (http://slidebase.binf.ku.dk/human_enhancers/presets).



TWAS

Two approaches

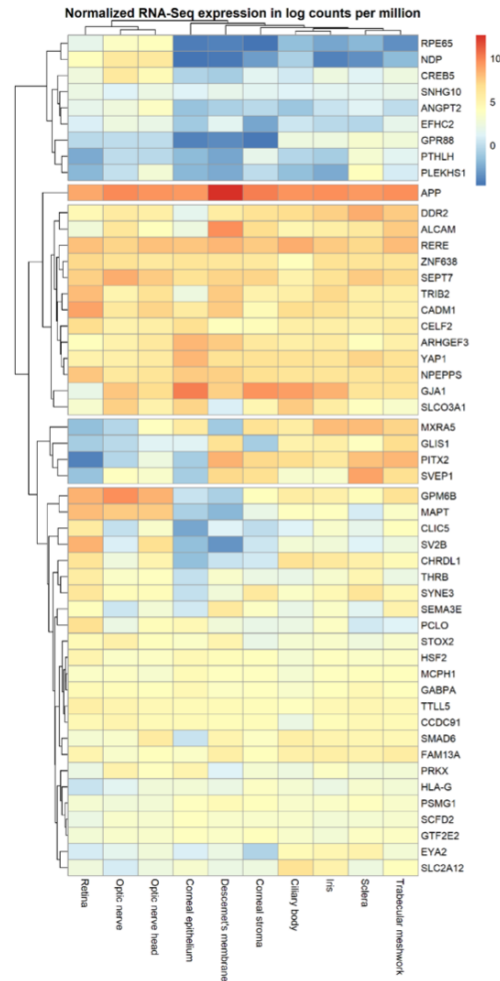
- MetaXcan
 - “we used our POAG cross-ancestry meta-analysis statistics, RNA-seq and genotype data from peripheral retina (EyeGEx) and 44 GTEX tissues”
 - “we identified 100 significant genes, which were selected as the most likely causal genes based on the integration of eQTL data”
- Summary Mendelian randomization
 - “we integrated our GWAS meta-analysis summary statistics with eQTL data from blood and retina”
- “Although the majority of the risk loci identified through the MetaXcan and SMR approaches were also identified in the meta-analysis, these analyses help with prioritizing the most likely functionally relevant genes”
 - “we used the approach implemented in FOCUS v0.5, a probabilistic framework that assigns a posterior probability for each gene causally driving TWAS associations in multiple tissues”

Differential expression of the previously unknown risk loci in eye tissues

Clustering analysis shows that the majority of these genes were expressed in eye tissues

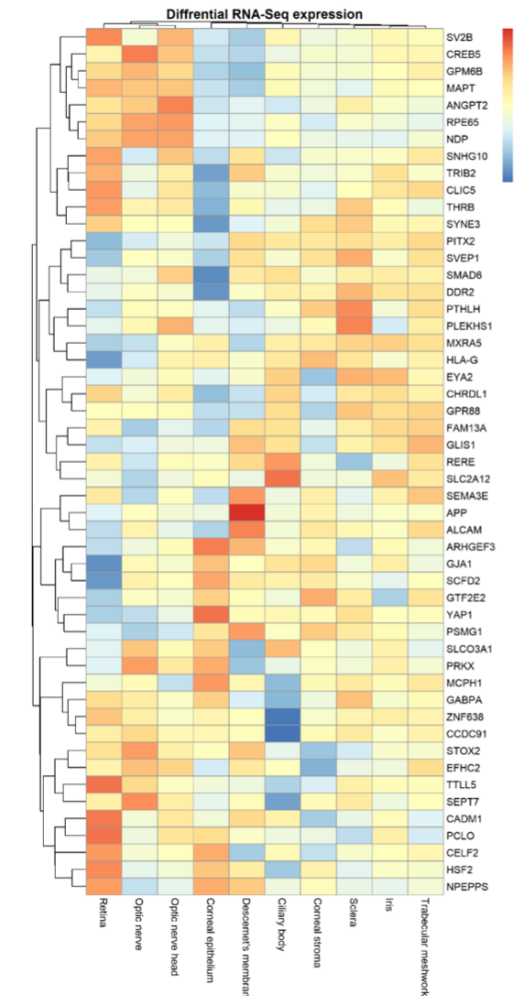
Supplementary Figure 7. Gene expression heatmaps in eye tissues.

a) Expression of the nearest genes to the most significant previously unknown risk SNPs.



We examined the differential expression of the previously unreported genes in ocular tissues likely to be involved in POAG pathogenesis, namely trabecular meshwork, ciliary body and optic nerve head, and found 36/51 (71%) of the genes differentially expressed in these tissues compared to the other eye tissues tested in this study

b) Differential gene expression of the nearest genes to the most significant previously unknown risk SNPs.



Drug targets (Open Targets)

“At least 16 of the POAG risk genes (nearest to the lead SNPs) are targeted by existing drugs, some of which are already in use/clinical trials for several eye or systemic diseases. The functional relevance of 14 of these 16 drug target genes is supported by the bioinformatic functional analyses we used in this study (i.e., eQTL, chromatin interaction, etc; Supplementary Data ”

Summary

“Integrating data from several lines of evidence described above, as well as the cross-ancestry fine-mapping and genetic pathways, provided support for specific genes potentially influencing POAG risk particularly *RERE*, *VCAM1*, *ZNF638*, *CLIC5*, *SLC2A12*, *YAP1*, *MXRA5*, and *SMAD6*.”

“For example, rs3777588, a lead GWAS SNP in this study, is an intronic variant within *CLIC5*. The CADD score for this variant is 16.58, providing support for the pathogenicity of this variant (determined by a CADD score > 12.37). The lead SNP is also an eQTL for *CLIC5* in both GTEx and retina, and gene-based analysis by incorporating eQTL data supported the involvement of *CLIC5* in POAG risk.”

“This approach also helped to select best genes near the lead SNPs or to shift the focus from the nearest genes to genes further away. For example, rs12846405, a lead GWAS SNP in this study, is an intergenic variant located between *MXRA5* and *PRKX*. Based on integration of eQTL data and gene-based analysis, *MXRA5* was prioritized as the most likely causal gene in this locus.”